Medical Image Segmentation in a Multiple Labelers Context: Application to the Study of Histopathology

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Segmentación de Imágenes Médicas en un Contexto de Múltiples Anotadores: Aplicación al Estudio de Histopatologías

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ABSTRACT

Breast cancer remains one of the deadliest diseases affecting women worldwide, making accurate diagnosis and treatment planning crucial for improving patient outcomes. Semantic segmentation of histopathology images plays a vital role in this process, as it enables precise identification and analysis of tissue structures. However, the current approaches face significant challenges in real-world scenarios where multiple annotators with varying expertise and reliability contribute to the labeling process. This crowdsourcing paradigm, while cost-effective, introduces inconsistencies that can compromise the quality of the segmentation results.

This thesis addresses two fundamental challenges in medical image segmentation: the inherent uncertainty in multiple annotator scenarios and the need for robust segmentation models that can handle varying image quality and annotator expertise. These challenges arise from the complex nature of histopathology images, where tissue structures can be ambiguous, and the varying levels of expertise among annotators lead to inconsistent labeling patterns. Additionally, the high cost and time requirements for expert annotations create a need for efficient solutions that can work with crowdsourced data.

To address these challenges, this thesis proposes a comprehensive approach that combines novel loss functions with robust deep learning architectures to handle multiple annotator scenarios effectively.

The first objective introduces a novel loss function that implicitly infers optimal segmentation while assessing labeler reliability without requiring explicit performance inputs. This approach differs from conventional methods by eliminating the need for direct supervision of labeler performance, instead using an intermediate reliability map that enables the model to prioritize information from reliable labelers while disregarding noisy labels. The mathematical foundation of this approach lies in the probabilistic modeling of annotator reliability across the input space.

The second objective focuses on the development of a robust segmentation architecture that combines the strengths of U-shaped deep learning models with the proposed reliability-aware loss function. This combination enables the model to learn both the segmentation task and the reliability patterns simultaneously, leading to more accurate and consistent results. The architecture's key innovation lies in its ability to adapt to varying image quality and annotator expertise levels without requiring explicit performance metrics.

The third objective involves validating the proposed approach through extensive experimentation on both synthetic datasets and real-world histopathology images. The evaluation demonstrates superior performance compared to state-of-the-art approaches in terms of segmentation accuracy and uncertainty quantification. The results show that the proposed method can effectively handle inconsistent annotations while maintaining high segmentation accuracy, making it particularly valuable for clinical applications.

The research makes significant contributions to the field by providing a robust solution for handling inconsistent annotations in medical image segmentation, particularly valuable in histopathology where expert annotations are expensive and time-consuming to obtain. The proposed approach shows promising results in reducing annotation costs while maintaining high segmentation accuracy, making it particularly relevant for clinical applications where precise segmentation of tissue structures is crucial for diagnosis and treatment planning. This work opens

new research directions in the areas of crowdsourced medical image analysis, uncertainty quantification in deep learning, and the development of more efficient annotation protocols for medical imaging.

Keywords: Crowdsourcing, Multiple Annotators, Histopathology, Breast Cancer, Semantic Image Segmentation, Gaussian Processes, Deep Learning, Medical Image Analysis

RESUMEN

El cáncer de mama sigue siendo una de las enfermedades más mortales en la población femenina en todo el mundo, lo que hace que el diagnóstico preciso y la planificación del tratamiento sean cruciales para mejorar las condiciones de los pacientes. La segmentación semántica de imágenes histopatológicas juega un papel vital en este proceso, ya que permite la identificación y análisis preciso de estructuras tisulares. Sin embargo, los enfoques actuales enfrentan desafíos significativos en escenarios del mundo real donde múltiples anotadores con diferentes niveles de experiencia y confiabilidad contribuyen al proceso de etiquetado. Este paradigma de crowdsourcing, aunque rentable, introduce inconsistencias que pueden comprometer la calidad de los resultados de segmentación.

Esta tesis aborda dos desafíos fundamentales en la segmentación de imágenes médicas: la incertidumbre inherente en escenarios de múltiples anotadores y la necesidad de modelos de segmentación robustos que puedan manejar la calidad variable de las imágenes y la experiencia de los anotadores. Estos desafíos surgen de la naturaleza compleja de las imágenes histopatológicas, donde las estructuras tisulares pueden ser ambiguas, y los diferentes niveles de experiencia entre los anotadores conducen a patrones de etiquetado inconsistentes. Además, el alto costo y los requisitos de tiempo para las anotaciones de expertos crean la necesidad de soluciones eficientes que puedan trabajar con datos crowdsourced.

El primer objetivo introduce una función de pérdida novedosa que infiere implícitamente la segmentación óptima mientras evalúa la confiabilidad de los etiquetadores sin requerir entradas explícitas sobre su rendimiento. Este enfoque difiere de los métodos convencionales al eliminar la necesidad de supervisión directa del rendimiento de los etiquetadores, utilizando en su lugar un mapa de confiabilidad intermedio que permite al modelo priorizar la información de etiquetadores confiables mientras descarta etiquetas ruidosas. La base matemática de este enfoque radica en el modelado probabilístico de la confiabilidad de los anotadores a través del espacio de entrada.

El segundo objetivo se centra en el desarrollo de una arquitectura de segmentación robusta que combina las fortalezas de los modelos de aprendizaje profundo en forma de U con la función de pérdida propuesta que tiene en cuenta la confiabilidad. Esta combinación permite que el modelo aprenda tanto la tarea de segmentación como los patrones de confiabilidad simultáneamente, lo que lleva a resultados más precisos y consistentes. La innovación clave de la arquitectura radica en su capacidad para adaptarse a diferentes niveles de calidad de imagen y experiencia de los anotadores sin requerir métricas de rendimiento explícitas.

El tercer objetivo valida el enfoque propuesto a través de experimentación exhaustiva tanto en conjuntos de datos sintéticos como en imágenes histopatológicas del mundo real. La evaluación demuestra un rendimiento superior en comparación con los enfoques más avanzados en términos de precisión de segmentación y cuantificación de incertidumbre. Los resultados muestran que el método propuesto puede manejar efectivamente anotaciones inconsistentes mientras mantiene una alta precisión de segmentación, lo que lo hace particularmente valioso para aplicaciones clínicas.

La investigación hace contribuciones significativas al campo al proporcionar una solución robusta para manejar anotaciones inconsistentes en la segmentación de imágenes médicas, particularmente valiosa en histopatología donde las anotaciones de expertos son costosas y requieren mucho tiempo para obtenerlas.

El enfoque propuesto muestra resultados prometedores en la reducción de costos de anotación mientras mantiene una alta precisión de segmentación, lo que lo hace particularmente relevante para aplicaciones clínicas donde la segmentación precisa de estructuras tisulares es crucial para el diagnóstico y la planificación del tratamiento. Este trabajo abre nuevas direcciones de investigación en las áreas de análisis de imágenes médicas crowdsourced, cuantificación de incertidumbre en aprendizaje profundo y el desarrollo de protocolos de anotación más eficientes para imágenes médicas.

Palabras clave: Crowdsourcing, Múltiples Anotadores, Histopatología, Cáncer de Mama, Segmentación Semántica de Imágenes, Procesos Gaussianos, Aprendizaje Profundo, Análisis de Imágenes Médicas

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ABBREVIATIONS

CAD Computer-Aided Diagnosis 2, 5, 8

CCGP Correlated Chained Gaussian Processes 18

CCGPMA Correlated Chained Gaussian Processes for Multiple Annotators xxii, 18, 19, 53, 55–57, 59, 93

CE Cross Entropy 62, 65

CGP Chained Gaussian Processes 18, 45, 46

CKA Centered Kernel Alignment 44

CNN Convolutional Neural Networks 3, 15, 67, 93

CR Crowdsourcing 16, 17

CT Computed Tomography 1, 2, 12

ELBO Evidence Lower Bound 19

EM Expectation-Maximization 15, 20

GCE Generalized Cross Entropy 62, 65, 86

GCECDL Generalized Cross-Entropy-based Chained Deep Learning 21

GP Gaussian Process 46, 58

GPMC Gaussian Process Monte Carlo 48

GPR Gaussian Process Regression 48

ISS Image Semantic Segmentation 2, 5, 8, 10, 14, 21, 23, 37, 91, 93

KAAR Kernel Alignment-Based Annotator Relevance Analysis 43, 44, 46, 57

LF Latent Function 18

LKAAR Localized Kernel Alignment-Based Annotator Relevance Analysis 44–46, 57

MAE Mean Absolute Error 62, 65

MITs Medical Imaging Techniques 1

ML Machine Learning 10, 12, 21

MRI Magnetic Resonance Imaging 1, 2

MV Majority Voting 10, 12, 20, 22

OCR Optical Character Recognition 12

PET Positron Emission Tomography 14

RCDNN Regularized Chained Deep Neural Network 47, 57

ROI Region of Interest 2, 8, 16, 31, 82

SGPMC Sparse Gaussian Process Monte Carlo 48

SGPR Sparse Gaussian Process Regression 48

SLFM Semi-Parametric Latent Factor Model 18, 51, 52

SS Semantic segmentation 3, 67

STAPLE Simultaneous Truth and Performance Level Estimation 13–15, 38

WSI Whole Slide Imaging xxii, 1, 5, 8, 16, 29, 31, 40, 82

CHAPTER

ONE

INTRODUCTION

1.1 Motivation

Since Roentgen's discovery of X-rays in 1895, medical imaging has advanced significantly, with modalities like radionuclide imaging, ultrasound, Computed Tomography (CT), Magnetic Resonance Imaging (MRI), and digital radiography emerging over the past 50 years. Modern imaging extends beyond image production to include processing, display, storage, transmission and analysis. [Zhou et al., 2021]. Other Medical Imaging Techniques (MITs) have arose during the last decades, some of them implying only the examination of certain pieces or tissues instead of complete patients, like histopathological images, which are images of tissue samples obtained from biopsies or surgical resections and are widely used for the diagnosis of diseases like cancer through Whole Slide Imaging (WSI) scanners [Rashmi et al., 2021].

Along with the advances in technologies for medical images acquisition, computational technologies on pattern recognition and artificial intelligence have also emerged, allowing the development of Computer-Aided Diagnosis (CAD) systems based on Machine Learning algorithms. These systems aim to assist physicians in the diagnosis and treatment of diseases, by providing a second opinion or by automating the analysis of medical images. [Panayides et al., 2020]. One of the most common tasks in which machine learning technologies are being used in the universe of medical images is Image Semantic Segmentation (ISS), which consists of assigning a label to each pixel in an image according to the object it belongs to. This is crucial for the development of CAD systems, as it allows the identification of Region of Interest (ROI) in the images, which can be used to detect and classify diseases [Azad et al., 2024].

The application of Machine Learning in medical imaging has grown significantly, including classification, segmentation, anomaly detection, super-resolution, image registration, and synthetic image generation [Brito-Pacheco et al., 2025]. Among imaging modalities, X-rays and CT scans are widely used for classification and anomaly detection, especially in pulmonary and oncological applications. MRI and ultrasound play a crucial role in segmentation and resolution enhancement, while PET/SPECT imaging is essential for anomaly detection in oncology and neurodegenerative diseases [Brito-Pacheco et al., 2025]. Histopathology is rapidly gaining prominence, particularly in segmentation and feature extraction, where AI-driven techniques aid in automated cancer diagnosis and tissue structure analysis. The integration of Deep Learning in histological image processing is revolutionizing pathology, enabling more precise and efficient diagnostics. A brief comparison of the tasks and medical image types based on counted referenced terms occurrences from recent literature review, can be seen in Figure **1-1**. [Yu et al., 2025], [Brito-Pacheco et al., 2025], [Ryou et al., 2025], [Hu et al., 2025], [Elhaminia et al., 2025]

Throughout the time that medical images and computational techniques have been developed, a coevolution has occurred, where the complexity of the tasks and the capabilities of the computational techniques have been increasing [Zhou et al., 2021]. Initially, these needs were covered with simple morphological

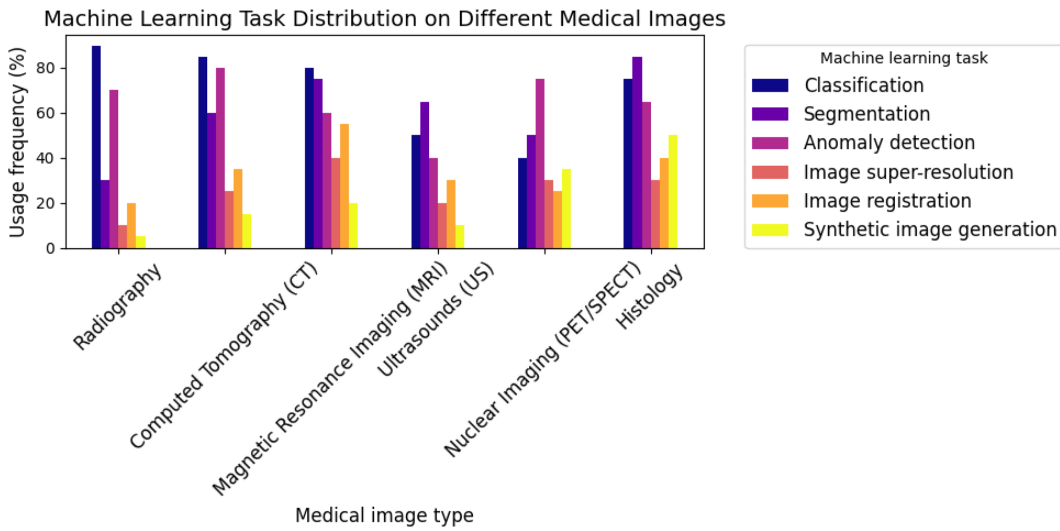


Figure **1-1** Estimation of the popularity of tasks and medical image types based on recent literature review (count of referenced terms.)

transformations, which implied no training process or elaborated optimization, this is an advantage in terms of development costs but has limited performance. However, as the complexity of the tasks increased, the need for more sophisticated techniques arose, leading to the application of advanced statistical tools and Machine Learning algorithms like Support Vector Machines, Decision Trees, and SGD Neural Networks [Avanzo et al., 2024]. The coevolution of advances in medical image acquisition, computational power (i.e. Moore's law) and statistical/mathematical techniques have led to a convergence for merging state of the art algorithms with medical imaging [Shalf, 2020]. Figure 1-2 shows a timeline of coevolution between some conspicuous advances in computational pattern recognition and its medical applications in different scopes (besides medical imaging) [Avanzo et al., 2024].

Convolutional Neural Networks (CNN) have been widely used in Semantic segmentation (SS), as they have outperformed traditional Machine Learning algorithms both medical and non medical images [Xu et al., 2024] [Sarvamangala and Kulkarni, 2022]. However, most CNN architectures are deep, which introduces the requirement of large amounts of training data. This introduces a

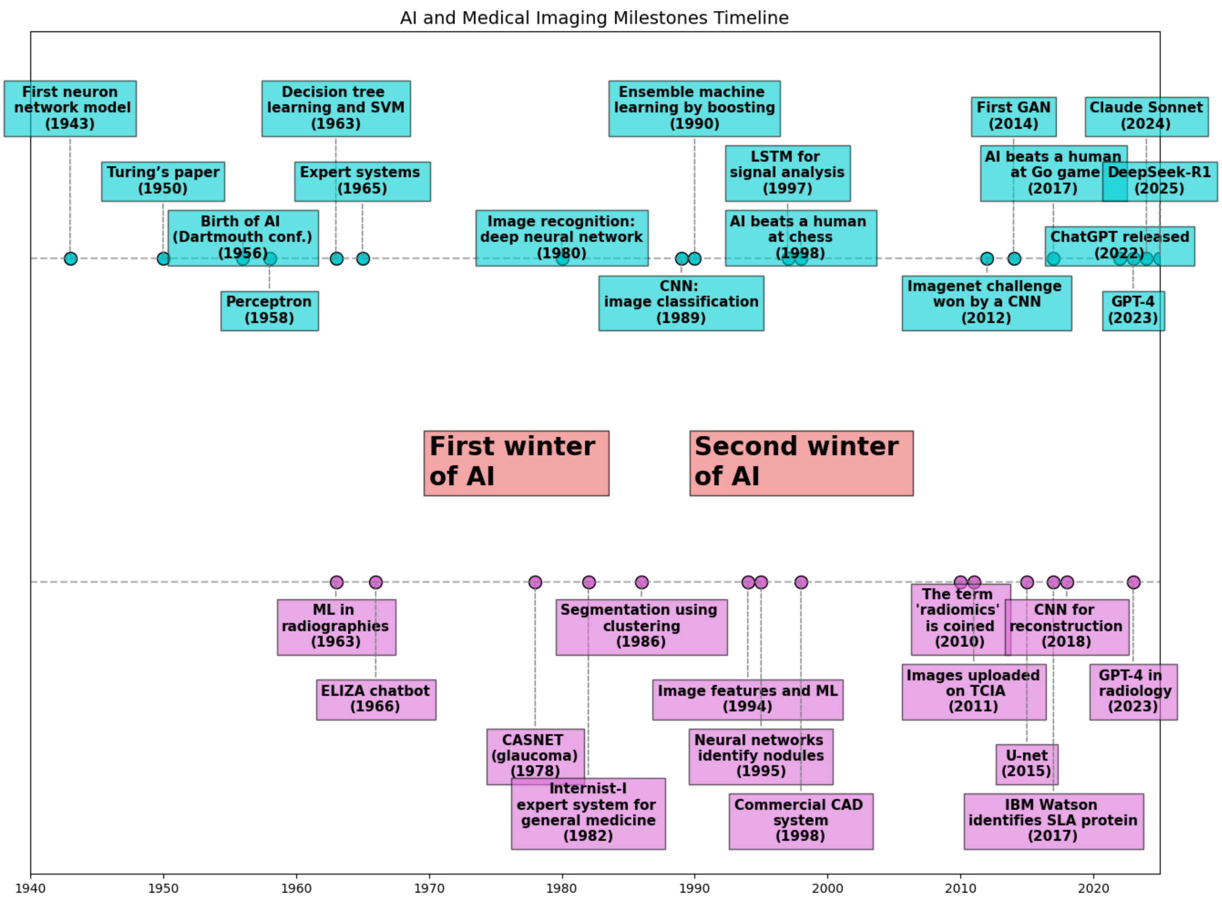


Figure **1-2** AI and Machine Learning in medical imaging recent timeline.

problem since the acquisition and annotation of medical images are expensive and time-consuming processes. This is especially true for ISS tasks, as they require pixel-level annotations, which is taxing in terms of cost, time and the logistics involved [Bhalgat et al., 2018]. Other modalities face this problem through less expensive annotation strategies like bounding boxes or anatomical landmarks for being used in a semi-supervised strategy [Shah et al., 2018].

However, many medical images datasets contain a high variability in class sizes and variations in colors, which is specially noticeable in histopathological images because of the usage of different staining techniques and other factors which can affect the color of the images (see section 2.2.3). This variability can lead to a significant loss of efficiency of Machine Learning models when using a mixed supervision strategy, as the model can be biased towards the most common classes or colors in the dataset [Shah et al., 2018].

This is were other solutions arise to tackle the problem of the weak image annotation while maintaining low costs. One of these solutions is the *crowdsourcing* strategy, which consists of having multiple annotators labeling the same image, and then combining the labels to obtain a consensus label [Lu et al., 2023]. This strategy can lead to a labeling cost reduction when different levels of expertise are combined, since the crowd may be composed of both experts and laymen, the latter being less expensive to hire [López-Pérez et al., 2023].

Recently, diagnosis, prognosis and treatment of cancer have heavily relied on histopathology, where tissue samples are obtained through biopsies or surgical resections and critical information that helps pathologists determine the presence and severity of malignancies [López-Pérez et al., 2024]. The segmentation of histopathological images enables precise identification of structures such as nuclei, glands, and tumors, which are essential for assessing disease progression and treatment response [Rashmi et al., 2021]. Accurate segmentation is particularly crucial in digital pathology, where whole-slide images (WSI) are analyzed using AI-powered CAD systems to support clinical decision-making [López-Pérez et al., 2024].

A major challenge in histopathological image segmentation arises from the variability in annotations provided by different pathologists. Unlike natural images, where object boundaries are often well-defined, histological structures may have ambiguous borders, leading to inconsistencies among annotators [López-Pérez et al., 2023]. Because of this, crowdsourcing labeling is one of the most popular approaches, as illustrated in Figure 1-3, [1]. These discrepancies highlight the need for models that can handle annotation uncertainty effectively. Leveraging crowdsourcing strategies and Machine Learning techniques that infer annotator reliability can enhance segmentation performance while reducing costs.

[1] obtained from a real world Triple Negative Breast Cancer (TNBC) dataset published in [López-Pérez et al., 2023]

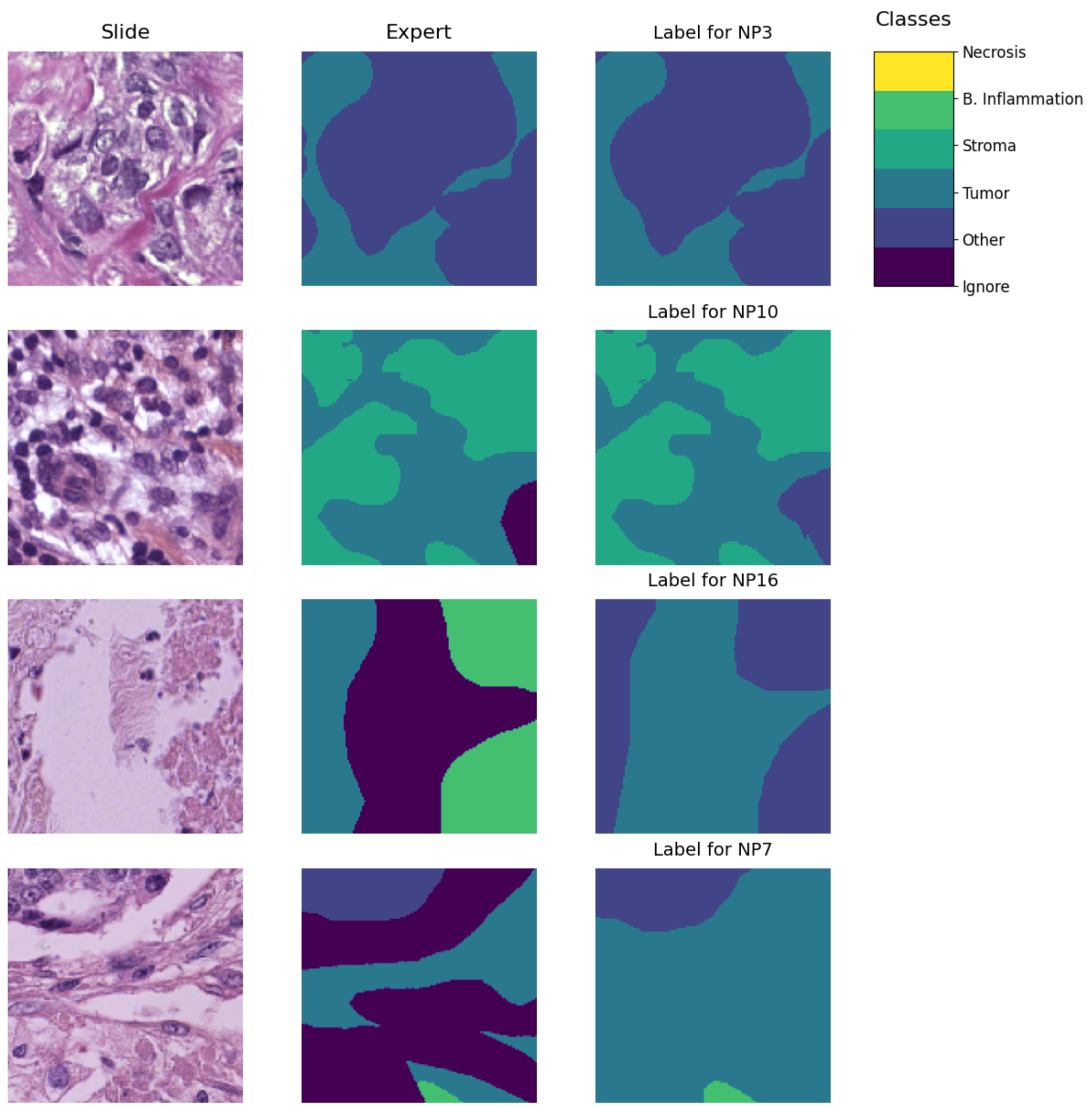


Figure **1-3** Example of a histopathological image segmented by multiple annotators, illustrating variations in label assignment.

1.2 Problem Statement

Throughout the development of medical technology and CAD, the task of ISS has become a crucial step in delivering precise diagnosis and treatment planning [Giri and Bhatia, 2024]. Particularly, in the area of histopathological studies, the usage of Whole Slide Images (WSI) is rather common since this method delivers high quality imaging and allows for the diagnosis of diseases like cancer [Lin et al., 2024].

ISS task consists of assigning a label to each pixel in an image according to the object it belongs to. Accurate segmentation is essential for the development of CAD systems, as it allows the identification of regions of interest (ROI) in the images, which can be used to detect and classify diseases and hence, treatment planning [Sarvamangala and Kulkarni, 2022]. However, modern computational solutions for ISS tasks involve the use of deep learning, which mostly rely large amounts of labeled data to train the models on supervised learning techniques. This means that the model is trained on a dataset with ground-truth labels, which are assumed to be correct and consistent across all samples. In practice, this assumption is often violated due to the high technical complexity of labeling these segments [2].

The process of labeling medical images is often managed with the help of specialized software tools that allow the annotators to draw the regions, delivering a standard format for the labeled masks [Habis, 2024]. Despite the help of these tools, the labeling process in WSI can have high costs, as it requires long hours of work from specialized personnel. In the pursuit of reducing costs, the labeling processes is often done by multiple labelers with varying levels of expertise, equalizing the cost of the labeling process. However, this strategy can lead to inconsistent labels, as the consensus between the labelers may not be exact due to the diversity in depth of knowledge and experience of the labelers [Xu et al., 2024]. These inconsistencies are discussed in the subsections 1.2.1 and 1.2.2.

[2] compared to a more trivial task like image classification on ordinary an well known classes like MNIST

1.2.1 Variability in Expertise Levels

One of the primary sources of inter-observer variability in medical image segmentation is the difference in expertise levels among annotators [López-Pérez et al., 2023]. Experienced radiologists and pathologists tend to produce highly precise annotations, whereas novice labelers may introduce systematic biases due to their limited familiarity with subtle image features. Studies have demonstrated that annotation accuracy *tends* to improve with experience, yet medical institutions often rely on a mix of annotators to manage costs and workload distribution [Lu et al., 2023].

The training background of annotators and institutional guidelines play a crucial role in shaping labeling practices. Different medical schools and hospitals may adopt distinct segmentation protocols, leading to inconsistencies when datasets are combined from multiple sources [López-Pérez et al., 2023]. For example, some institutions may emphasize conservative delineation of tumor boundaries, while others adopt a more inclusive approach. Such variations contribute to systematic biases in medical image datasets [Banerjee et al., 2025].

Medical images frequently contain complex structures which are difficult to distinguish, making segmentation inherently subjective. For instance, tumor margins in histopathological slides may not have well-defined edges, leading to variations in how different annotators delineate the regions of interest [Carmo et al., 2025]. These discrepancies arise not only from technical expertise but also from differences in perception and interpretation.

1.2.2 Technical Constraints and Image Quality

Technical constraints in medical imaging, such as resolution differences, noise levels, and contrast variations, can significantly impact segmentation accuracy.

Lower-resolution images may obscure fine structures, leading to inconsistencies in boundary delineation [Zhou et al., 2024].

When combined with long sessions, images with insufficient quality might also increase the cognitive load of the annotators, leading to fatigue and reduced precision in labeling [Kim et al., 2024]. This is particularly relevant in histopathological studies, where the staining process and tissue preparation can introduce color variations and artifacts that affect image quality, even if the same scanning equipment is used [Karthikeyan et al., 2023].

1.2.3 Research Question

Given the challenges posed by inconsistent labels in medical image segmentation, this work aims to address the following research question:

> **Research Question**
>
> *How can we develop a learning approach for ISS tasks in medical images that can adapt to inconsistent labels without requiring explicit supervision of labeler performance, while addressing challenges related to variability in expertise levels and technical constraints, and maintaining interpretability and generalization?*

1.3 State of the Art

Certainly, in general Machine Learning (ML) classification tasks [3] where multiple annotators are involved, Majority Voting (MV) is by far the simplest possible approach to implement. This concept was born multiple times and divergently in

[3] In this work, image segmentation is considered as a particular case of classification in which target classes are assigned pixel-wise.

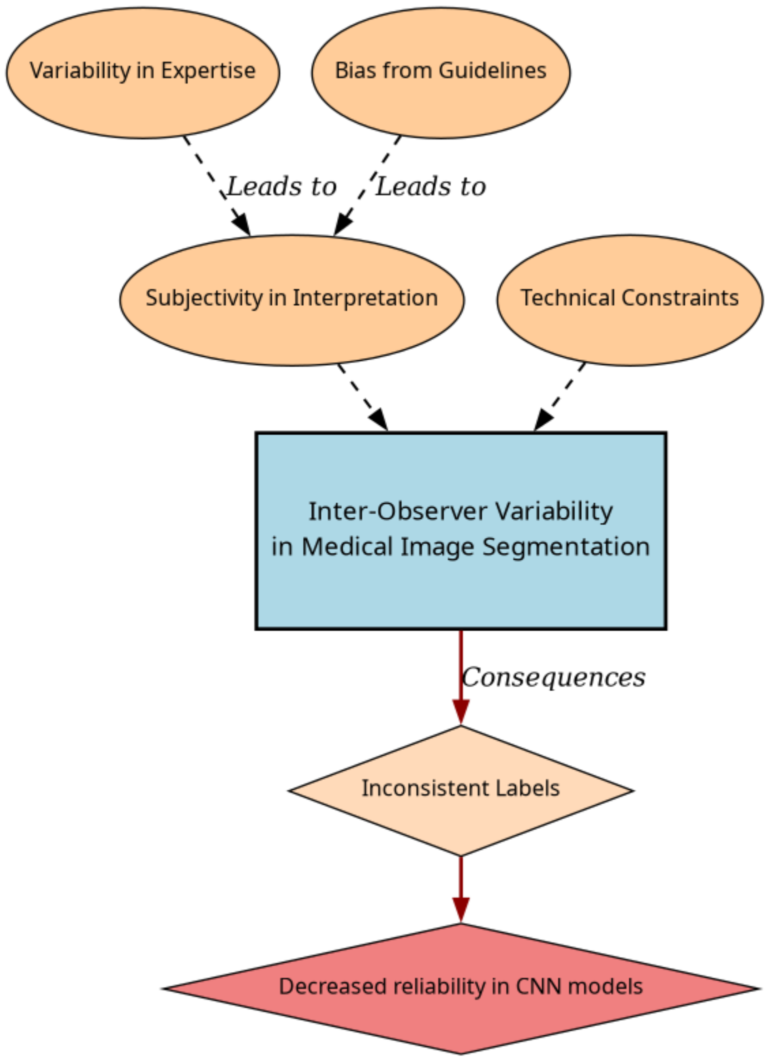


Figure **1-4** Summary diagram for problem Statement multiple fields, but it was described as relevant for ML and pattern recognition labeling for classification in [Lam and Suen, 1997], in which the approach is exposed as simple, yet powerful. The authors describe the MV as a method that can be used to improve the accuracy of classification tasks by combining the labels of multiple annotators. The method is based on the assumption that the majority vote of the annotators is more likely to be correct than the vote of a single annotator. The authors also describe the method as a straightforward way to improve the accuracy of classification tasks without the need for complex algorithms or additional data. The authors also prove this method to deliver very similar results to more complicated approaches (Bayesian, logistic regression, fuzzy integral, and neural network) in the particular task of Optical Character Recognition (OCR). Despite its simplicity, modern solutions for delivering accurate medical image segmentation models still rely on Majority Voting at some stage. In [Elnakib et al., 2020], a majority voting strategy is used for delivering a final output based on the labels of multiple models (VGG16-Segnet, Resnet-18 and Alexnet) in CT images for Liver Tumor Segmentation. Majority voting as a technique for label aggregation is a powerful approach for its simplicity in many use cases in which the target to be labeled is not tied to an expertise related task, otherwise, the assumption of equal expertise among the labelers may introduce bias in the final label, which is not desirable in the case of highly technical annotations like medical images. In subsection 1.3.1, a review is done on literature which no longer assumes the naive approach of equal expertise among labelers and face the challenge of learning from inconsistent labels will be presented.

1.3.1 Facing Annotation Variability in Medical images

Learning from crowds approaches in general face the challenge of not having a ground truth label and hence, an intrinsic difficulty in measuring the real reliability of the labelers annotations. Some approaches assume beforehand a certain level of expertise for each labeler based on experience as an input, like in [TIAN and Zhu, 2015], which introduce the concept of max margin majority voting, using the reliability vector as weights for the binary and multiclass classifier. The crowdsourcing margin is the minimal difference between the aggregated score of the potential true label and the scores for other alternative labels. Accordingly, the annotators' reliability is estimated as generating the largest margin between the potential true labels and other alternatives. The problem introduced in this approach is assuming an stationary reliability per expert across the whole input space, which is imprecise since annotators performance may change between different tasks or even between different regions of the same image.

STAPLE Mechanism

The Simultaneous Truth and Performance Level Estimation (STAPLE) algorithm, introduced in [Warfield et al., 2004] is a probabilistic framework that estimates a hidden true segmentation from multiple segmentations provided by different raters. It also estimates the reliability of each rater by computing their sensitivity and specificity.

The STAPLE algorithm's goal is to maximize the log likelihood function:

$$(\hat{\mathbf{p}}, \hat{\mathbf{q}}) = \arg\max_{\mathbf{p},\mathbf{q}} \ln f(\mathbf{D}, \mathbf{T} \mid \mathbf{p}, \mathbf{q}). \tag{1-1}$$

Where $\mathbf{D}$ is the set of segmentations provided by the raters, $\mathbf{T}$ is the hidden true segmentation, p is the sensitivity and q is the specificity of the raters.

This is achieved by using the Expectation-Maximization algorithm to maximize the log likelihood function, which is done iteratively with step computations:

$$
\begin{aligned}
(p_j^{(k)}, q_j^{(k)}) = \arg\max_{p_j, q_j} & \sum_{i:D_{ij}=1} W_i^{(k-1)} \ln p_j \\
& + \sum_{i:D_{ij}=1} \left(1 - W_i^{(k-1)}\right) \ln(1 - q_j) \\
& + \sum_{i:D_{ij}=0} W_i^{(k-1)} \ln(1 - p_j) \\
& + \sum_{i:D_{ij}=0} \left(1 - W_i^{(k-1)}\right) \ln q_j .
\end{aligned}
\tag{1-2}
$$

The weight terms $W_i^{(k-1)}$ and $(1 - W_i^{(k-1)})$ represent the current estimates of the probability that voxel i belongs or does not belong to the true segmentation from the previous E-step. The log-likelihood contributions $\ln p_j$, $\ln(1-q_j)$, $\ln(1-p_j)$, and $\ln q_j$ account for true positives, false positives, false negatives, and true negatives, respectively, weighted by the current estimate of true segmentation probabilities.

The capacity of STAPLE to accurately estimate the true segmentation, even in the presence of a majority of raters generating correlated errors, was demonstrated, which makes it theoretically a strong choice for setting a ground-truth in binary or multiclass medical ISS tasks.

The popularity and performance of STAPLE has led to its usage in modern applications medical image, 3d spatial images due to its assumption of decision space being based on voxel-wise decisions, like the authors in [Grefve et al., 2024] which applied the algorithm on Positron Emission Tomography (PET) images. Other authors still rely heavily on STAPLE for setting a ground truth consensus for histopathological images, like [Qiu et al., 2022].

However, the STAPLE algorithm has some limitations. It assumes independent rater errors, which may not hold in practice, leading to biased estimates. STAPLE is also sensitive to low-quality annotations, potentially degrading final segmentations if the weights are not initialized correctly. The algorithm tends to over-smooth results, blurring fine details, and struggles with multi-class segmentation. Computationally, it is expensive due to its iterative Expectation-Maximization (EM) approach. Additionally, STAPLE cannot correct systematic biases in annotations and depends on initial estimates, impacting accuracy. Lastly, the estimated performance levels lack interpretability, making it difficult to assess annotator reliability effectively.

Finally, while STAPLE proves valuable for aggregating existing labels and supporting other methodologies, its primary limitation lies in its inability to generate annotations for novel, unlabeled images. This restricts its applicability to scenarios where multiple annotations already exist, rather than serving as a standalone solution for new image segmentation tasks.

U-shaped CNNs

Since the introduction of U-Net [Ronneberger et al., 2015] in 2015 for biomedical image segmentation, U-shaped CNNs have become a prevalent architecture in medical image segmentation tasks. The U-Net's success stems from its ability to capture both global and local information through its contracting and expanding paths, making it particularly effective for complex and heterogeneous structures, even with limited annotated data. This architecture has been successfully applied to various medical image segmentation tasks, including organ segmentation, tumor segmentation, and brain structure segmentation [**?**].

The U-Net architecture consists of a symmetric encoder-decoder structure with skip connections. The encoder path progressively reduces spatial dimensions while increasing feature channels through a series of convolutional and max-pooling layers, capturing high-level semantic information. The decoder path uses transposed convolutions to gradually recover spatial resolution while reducing feature channels. Skip connections between corresponding encoder and decoder layers preserve fine-grained details by concatenating high-resolution features from the encoder with upsampled features in the decoder, enabling precise localization of structures.

In [López-Pérez et al., 2024] two networks are trained for delivering a final segmentation. One network is trained to estimate the annotators reliability and another one is trained to segment the image. The first network is a Deep Neural Network that takes as input features of image and the one-hot encoded identifiers of the labelers and outputs a reliability map across the image feature space. This map is then used to weight the contribution of each annotator to the final segmentation. The second network is the U-Net used for segmentation.

In this approach, it is assumed that the images are labeled for at least one labeler and not all of them, which is closer to a real world scenario, in which images with variability in the amount of annotations per patch is common. Hence, the input data can be modeled as:

$$\mathcal{D} = (\mathbf{X}, \tilde{\mathbf{Y}}) = \{(\mathbf{x}_n, \tilde{\mathbf{y}}_n^r) : n = 1, \ldots, N; r \in R_n\}, \tag{1-3}$$

Where every $\mathbf{x}_n$ is an input patch from a ROI in one WSI, $\tilde{\mathbf{y}}_n$ is the noisy annotation from the r labeler, N is the number of patches in the dataset and $R_n \subset \{1, \ldots, R\}$ is the set of labelers that annotated the image $\mathbf{x}_n$.

The authors then assume the annotator network to deliver a reliability map $\{\hat{\mathbf{A}}_\phi^{(r)}(\mathbf{x})\}_{r \in R_n}$ with different dimensions:

- Crowdsourcing (CR) global: a single reliability vector per labeler with dimensions C which represent global reliability of the labeler across all input space.
- CR image: a single reliability vector per image per labeler with dimensions C which represent local reliability of the labeler across the image.
- CR pixel: a reliability matrix per image per labeler, with dimensions C which represent local reliability of the labeler across all the pixels in the image.

These differences in dimensions are determined by the feature extraction space from segmentation network which feed the input of the annotator network, which the authors vary for experimentation purposes.

Let $\mathbf{p}_\theta(\mathbf{x}_n)$ be the estimated latent (ground truth) segmentation produced by the segmentation U-Net network. The estimated segmentation probability mask for each annotator is then computed as the product:

$$\mathbf{p}^{(r)}_{\theta,\phi}(\mathbf{x}_n) := \mathbf{A}^{(r)}_\phi(\mathbf{x}) \odot \mathbf{p}(\mathbf{x}_n), \tag{1-4}$$

where $\odot$ is the element-wise product and ϕ and θ are the parameters of the annotator network and the segmentation UNet network, respectively, being the latter initialized with a ResNet34 backbone pre-trained on ImageNet.

The authors propose a loss function involving cross-entropy and a trace based regularization on the reliability map, originally proposed in [Zhang et al., 2020] which combined, looks like:

$$\mathcal{L}(\theta,\phi) := \sum_{n=1}^{N}\sum_{r=1}^{R} \mathbb{I}\left(\tilde{\mathbf{y}}^{(r)}_n \in R_n\right) \cdot \left[\mathrm{CE}\left(\mathbf{A}^{(r)}_\phi(\mathbf{x}_n) \cdot \mathbf{p}_\theta(\mathbf{x}_n), \tilde{\mathbf{y}}^{(r)}_n\right) + \lambda \cdot \mathrm{tr}\left(\mathbf{A}^{(r)}_\phi(\mathbf{x}_n)\right)\right] \tag{1-5}$$

Being $\mathbb{I}$ the indicator function, CE the cross-entropy loss, and λ the regularization parameter.

When evaluated on a Triple Negative Breast Cancer dataset, this approach achieves a Dice coefficient of 0.7827, outperforming STAPLE (0.7039) and matching expert-supervised performance (0.7723). The CR image reliability modeling proved most effective, on the other hand CR pixel, while potentially offering finer-grained reliability estimation, requires significantly more training data.

Despite the proven performance of the approach by the authors, solving the problem of multiple labelers with two networks can be overwhelming for the optimization process, requiring large amounts of annotated data to properly codify the annotators spatial reliabilities, which could be managed by a single model with an appropriate loss function.

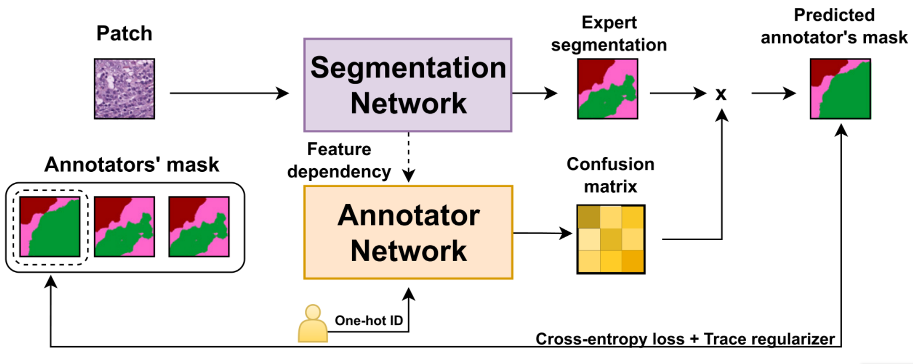


Figure **1-5** Proposed framework for the approach in [López-Pérez et al., 2024].

Bayesian Models

Bayesian approaches are a good choice for handling label noise and uncertainty in the labelers. In [Gil-González and Álvarez Meza, 2023] the authors propose a novel approach based on Gaussian Processes for modeling the relationship between the annotators' reliability and the input data, while also preserving the interdependencies among the annotators. This is achieved by introducing Correlated Chained Gaussian Processes for Multiple Annotators (CCGPMA), a framework based on the well known Chained Gaussian Processes (CGP). CGP on itself cannot consider inter-annotator dependencies, thus, the authors introduce the Correlated Chained Gaussian Processes (CCGP) thus, the authors introduce the CCGP to model correlations between the GP latent functions, which are supposed to be generated from a Semi-Parametric Latent Factor Model (SLFM):

$$f_j(\mathbf{x}_n) = \sum_{q=1}^{Q} w_{j,q}\mu_q(\mathbf{x}_n), \tag{1-6}$$

where $f_j : \mathcal{X} \to \mathbb{R}$ is a Latent Function (LF), $\mu_q(\cdot) \sim \mathcal{GP}(0, k_q(\cdot,\cdot))$ with $k_q : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ being a kernel function, and $w_{j,q} \in \mathbb{R}$ is a combination coefficient ($Q \in \mathbb{N}$). This leads to a joint distribution of the form:

$$p(\mathbf{y}, \hat{\mathbf{f}}, u|\mathbf{X}) = p(\mathbf{y}|\boldsymbol{\theta}) \prod_{j=1}^{J} p(\mathbf{f}_j|\mathbf{u})p(\mathbf{u}), \tag{1-7}$$

where **y** is the vector of noisy labels, $\hat{\mathbf{f}}$ is the vector of latent functions, u represents the inducing points, and **X** is the input data.

Combined with inducing-variables based methods for sparse GP approximations, and maximizing an Evidence Lower Bound (ELBO) for the estimation of the variational parameters, the authors reach a model whose variational expectations are not analytically tractable, and hence, the authors derive a Gaussian-Hermite quadrature approach.

Finally, the authors extend this approach for being applied to classification and regression, reaching the only known approach to involve chained gaussian processes in multiple annotators classification and regression tasks while preserving the interdependencies among the annotators, and also outperforming GPC-MV[4], MA-LFC-C[5], MA-DGRL[6], MA-GPC[7], MA-GPCV[8], MA-DL[9], KAAR[10].

CCGPMA on itself proposes a good approach for handling label noise and uncertainty in the labelers for regression and classification tasks, while also preserving the interdependencies among the annotators, however, it does not face the image segmentation problem, which is the main focus of this work. Besides, handling so many latent functions during the optimization process is computationally expensive, making it on itself infeasible for large and high resolution datasets.

[4] A GPC using the MV of the labels as the ground truth.
[5] A LRC with constant parameters across the input space.
[6] A multi-labeler approach that considers as latent variables the annotator performance.
[7] A multi-labeler GPC, which is an extension of MA-LFC.
[8] An extension of MA-GPC that includes variational inference and priors over the labelers' parameters.
[9] A Crowd Layer for DL, where the annotators' parameters are constant across the input space.
[10] A kernel-based approach that employs a convex combination of classifiers and codes labelers dependencies.

1.3.2 Facing Noisy Annotations and Low-Quality Data

The problem of low-quality data and noisy annotations has been tackled with various strategies. One such approach is the use of deep learning models that incorporate loss functions designed to mitigate the effects of unreliable labels. Traditional methods such as MV or EM have been widely used for aggregating multiple annotators' inputs. However, they assume a homogeneous reliability of annotators, which may not hold in real-world scenarios.

Loss Functions in Deep Learning Models

Loss functions are fundamental components in deep learning models that quantify how well a model's predictions match the ground truth. They serve as the objective function that guides the learning process by measuring the discrepancy between predicted and actual values. In classification tasks, the most common loss functions are Cross-Entropy (CE) and Mean Absolute Error (MAE) [Zhang and Sabuncu, 2018]. CE is particularly effective for classification as it heavily penalizes confident but wrong predictions, though it can be sensitive to noisy labels. MAE, on the other hand, is more robust to outliers and assigns equal weights to all mistakes, but typically requires more training iterations. For image segmentation tasks, specialized loss functions have been developed to handle the unique challenges of pixel-wise classification. The Dice loss, which measures the overlap between predicted and ground truth regions, is widely used in medical image segmentation [Zhao et al., 2020]. More recently, the Generalized Cross Entropy (GCE) loss has emerged as a robust alternative that combines the benefits of both CE and MAE, allowing for better handling of noisy labels through a tunable parameter that controls sensitivity to outliers. However, to the best of our knowledge, none of the current loss functions approaches have focused on the multiple labelers scenario, neither on the per labeler spatial reliability estimation.

Generalized Cross-Entropy for Multiple Annotators Classification

A more recent approach was proposed by [Triana-Martinez et al., 2023], introducing a Generalized Cross-Entropy-based Chained Deep Learning (GCECDL) framework. This method addresses the limitations of traditional label aggregation techniques by modeling each annotator's reliability as a function of the input data. The approach effectively mitigates the impact of noisy labels by using a noise-robust loss function, balancing Mean Absolute Error (MAE) and Categorical Cross-Entropy (CE). Unlike prior approaches, GCECDL accounts for the dependencies among annotators while encoding their non-stationary behavior across different data samples. Their experiments on multiple datasets demonstrated superior predictive performance compared to state-of-the-art methods, particularly in cases where annotations were highly inconsistent.

The strategy of the authors effectively unlocks the potential of ML models to handle low-quality data and noisy annotations, but it is bounded to classifications tasks only, not being by itself applicable to segmentation tasks. The TGCE equation for handling multiple annotators is defined as:

$$\mathrm{TGCE}(\mathbf{y}, f(\mathbf{x}); \tilde{\lambda}_x, \tilde{C}) = \tilde{\lambda}_x \frac{1 - (\mathbf{1}^\top(\mathbf{y} \odot f(\mathbf{x})))^q}{q} + (1 - \tilde{\lambda}_x)\frac{1 - (\tilde{C})^q}{q}, \tag{1-8}$$

where $\tilde{\lambda}_x$ represents the annotator reliability, $\tilde{C}$ is a constant, q is a parameter that controls the balance between MAE and CE behavior, $\mathbf{y}$ is the annotation vector, and $f(\mathbf{x})$ is the model prediction. This approach is more deeply discussed in chapter 4.

1.4 Aims

1.4.1 General Aim

The main purpose of this work is to adaptively infer the best possible segmentation in ISS tasks for medical images through a novel approach, without needing separate

Table **1-1** Summary of state-of-the-art approaches for handling multiple annotators in medical image segmentation

Approach	**Key Characteristics**	**Advantages**	**Limitations**
MV [Lam and Suen, 1997]	Simple aggregation of multiple annotators' labels	- Simple to implement - No complex algorithms required - Proven effective in many cases	- Assumes equal expertise - Sensitive to correlated errors - May introduce bias in expert tasks
STAPLE [Warfield et al., 2004]	Probabilistic framework estimating true segmentation and rater reliability	- Handles binary and multiclass segmentation - Estimates rater sensitivity/specificity - Robust to correlated errors	- Computationally expensive - Sensitive to low-quality annotations - Tends to over-smooth results - Limited interpretability
U-Net based approaches [Ronneberger et al., 2015]	Deep Learning architecture with encoder-decoder structure	- Captures both global and local information - Effective for complex structures - Works well with limited data	- Requires large amounts of training data - Complex optimization process - May need specialized loss functions
Bayesian Models (CCGPMA) [Gil-Gonzalez et al., 2023]	Gaussian Processes modeling annotator reliability	- Preserves annotator interdependencies - Handles label noise effectively - Works for classification and regression	- Computationally expensive - Not directly applicable to segmentation - Complex optimization process
Generalized Cross-Entropy (GCE) [Zhang and Sabuncu, 2018]	Loss function balancing MAE and CE	- Robust to noisy labels - Accounts for non-stationary behavior - Handles annotator dependencies	- Limited to classification tasks - Requires parameter tuning - May need adaptation for segmentation

inputs about the labeler while offering an interpretable reliability space per labeler and properly facing the performance inconsistency across the annotators space and the variability of images quality.

1.4.2 Specific Aims

- To develop a novel loss function for ISS tasks in medical images, which accurately infers the underlying ground truth segmentation mask without needing separate inputs about the labeler performance.

- To introduce a tensor map which codifies the reliability of each labeler based only on the inputs, allowing the model to implicitly weigh the labelers based on their performance across the mask and classes space.

- To develop and test a deep learning model for ISS tasks in medical images, which can learn from inconsistent labels and improve the segmentation performance compared to other solutions in state of the art.

CHAPTER

TWO

CONCEPTUAL PRELIMINARIES

2.1 Modern Concept of Digital Image

A digital image is a numerical representation of a visual scene, captured through various imaging devices and stored in a computer. From a mathematical perspective, a digital image can be represented as a function $f(x, y)$ that maps spatial coordinates (x, y) to intensity values. In the discrete domain, this function is sampled at regular intervals, creating a matrix of values known as pixels (picture elements).

2.1.1 Types of Digital Images

Grayscale Images

Grayscale images are the simplest form of digital images, where each pixel represents a single intensity value. Mathematically, a grayscale image can be represented as a 2D matrix I of size $M \times N$, where each element $I(i, j)$ represents the intensity at position (i, j). The intensity values typically range from 0 (black) to 255 (white) in 8-bit images, or from 0 to 65535 in 16-bit images.

Color images

Color images extend the grayscale concept by representing each pixel with multiple channels, typically Red, Green, and Blue (RGB). A color image can be represented as a 3D matrix I of size $M \times N \times 3$, where $I(i, j, k)$ represents the intensity of the k-th color channel at position (i, j). Other color spaces like HSV (Hue, Saturation, Value) or CMYK (Cyan, Magenta, Yellow, Key) are also commonly used in different applications.

Multispectral images

Multispectral images capture information across multiple wavelength bands beyond the visible spectrum. These images can be represented as a 3D matrix I of size $M \times N \times B$, where B is the number of spectral bands. Each band $I(i, j, b)$ represents the intensity at position (i, j) for the b-th spectral band. This representation is particularly useful in medical imaging, remote sensing, and several scientific applications.

3D Images and Volumetric Data

Three-dimensional images extend the concept of pixels to voxels (volume elements). A 3D image can be represented as a 3D matrix V of size $M \times N \times D$, where D represents the depth dimension. Each voxel $V(i, j, k)$ represents the intensity at position (i, j, k) in the 3D space. This representation is fundamental in medical imaging (CT, MRI), scientific visualization, and computer graphics.

2.1.2 Mathematical Representations

The mathematical foundation of digital images relies on several key concepts:

- **Sampling**: The process of converting a continuous image into a discrete representation. According to the Nyquist-Shannon sampling theorem, the sampling frequency must be at least twice the highest frequency present in the image to avoid aliasing.

- **Quantization**: The process of converting continuous intensity values into discrete levels. The number of quantization levels determines the image's bit depth and affects its quality and storage requirements.

- **Resolution**: The number of pixels per unit length in an image, typically measured in pixels per inch (PPI) or dots per inch (DPI).

- **Dynamic range**: The ratio between the maximum and minimum measurable light intensities in an image, often expressed in decibels (dB).

The mathematical representation of a digital image can be expressed as:

$$I(x, y) = \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} f(i, j) \cdot \delta(x - i, y - j) \tag{2-1}$$

where $I(x, y)$ is the digital image, $f(i, j)$ represents the intensity values, and $\delta(x - i, y - j)$ is the Kronecker delta function [Duarte and Baraniuk, 2012].

For color images, the representation extends to:

$$I(x, y) = \begin{bmatrix} I_R(x, y) \\ I_G(x, y) \\ I_B(x, y) \end{bmatrix} \tag{2-2}$$

where I_R, I_G, and I_B represent the red, green, and blue channels respectively.

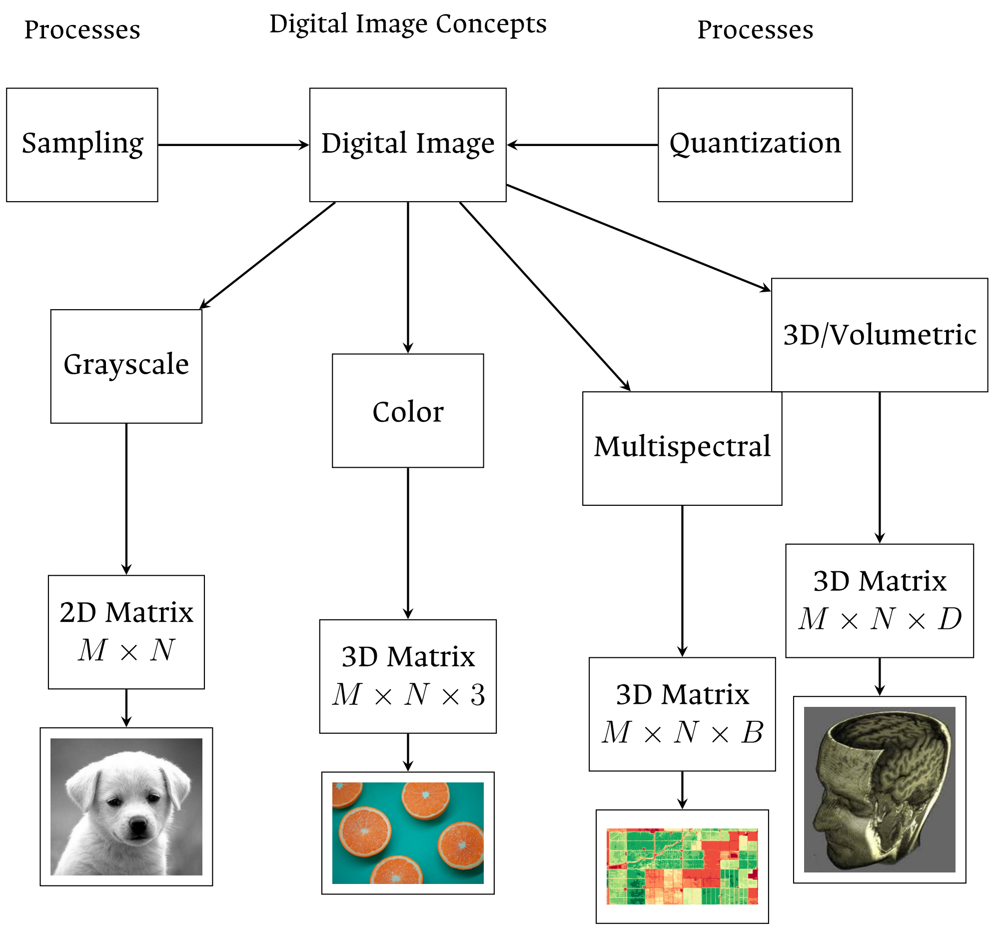


Figure **2-1** Overview of digital image concepts and their mathematical representations. The figure shows the main types of digital images (grayscale, color, multispectral, and volumetric), their mathematical representations, and the fundamental processes of sampling and quantization. Example images are included to illustrate each type.

2.2 Digital Histopathological Images

Digital histopathology represents a significant advancement in medical imaging, where traditional glass slides containing tissue samples are digitized using specialized scanning devices. This transformation has revolutionized the field of pathology by enabling remote diagnosis, computer-aided analysis, and digital archiving of tissue samples [Amgad et al., 2019]. This process has evolved significantly over the past few decades, as shown in Figure 2-2.

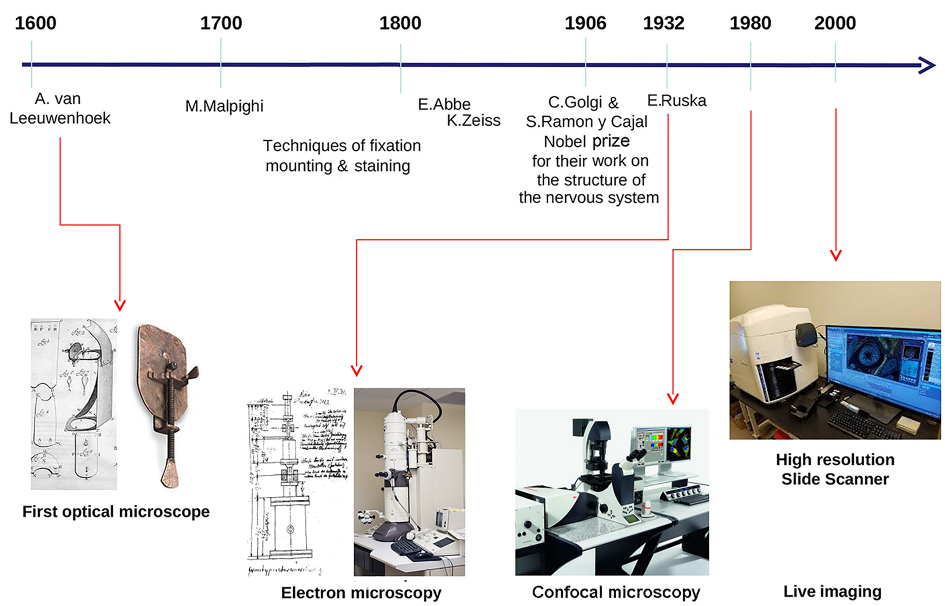


Figure 2-2 Histology evolution timeline. (Image from [Mazzarini et al., 2021]).

2.2.1 Whole Slide Imaging (WSI)

Whole Slide Imaging (WSI) is the process of digitizing entire glass slides at high resolution, creating a digital representation that can be viewed, analyzed, and shared electronically. Modern WSI scanners use sophisticated optical systems that capture multiple fields of view at high magnification, which are then stitched together to create a seamless digital image [Hu et al., 2025]. These systems incorporate high-resolution objectives with magnifications ranging from 20x to 40x, precise motorized stages for accurate slide positioning, automated focus systems to maintain image quality, and high-quality cameras equipped with large sensor arrays. The resulting digital slides can reach sizes of several gigabytes, containing billions of pixels that capture the microscopic details of tissue samples [Hu et al., 2025]. Figure 2-3 shows a whole slide imaging system by Omnyx for slide digitization and a comprehensive digital pathology interface from Omnyx designed to streamline pathologists' diagnostic workflow [Farahani et al., 2015].

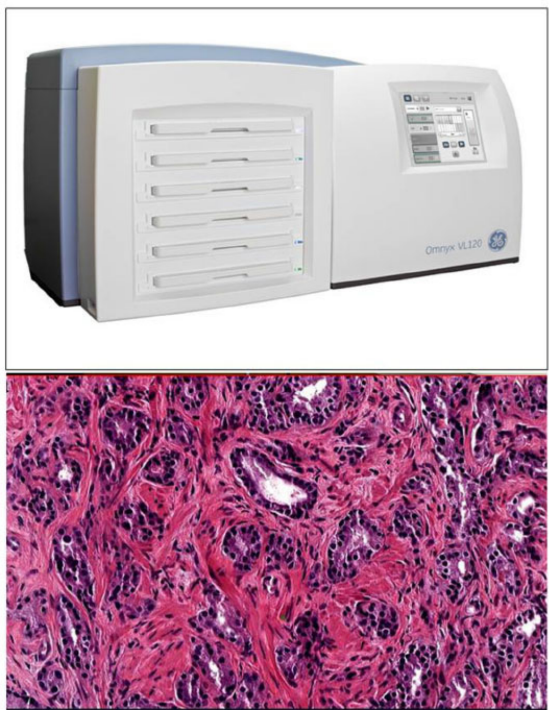


Figure **2-3** (Above) Whole slide imaging system by Omnyx for slide digitization. (Below) Comprehensive digital pathology scan visualization from Omnyx designed to streamline pathologists' diagnostic workflow. (From [Farahani et al., 2015]).

2.2.2 Regions of Interest (ROI)

In digital histopathology, Region of Interests (ROIs) are specific areas within a whole slide image that contain diagnostically relevant information. The importance of ROIs lies in their ability to focus computational analysis on relevant areas, reduce computational complexity in automated systems, facilitate targeted diagnosis and research, and enable efficient storage and transmission of critical information.

Selecting ROIs is a fundamental step in medical image analysis across all imaging modalities, where an ROI represents a subset of an image appropriate for the intended analysis and is typically identified manually by experts [Hossain et al., 2023]. In modern pathology, the analysis involves processing multidimensional and high-resolution WSI tiles automatically, containing an overwhelming quantity of structural and functional information. Despite recent improvements in computing capacity, analyzing such extensive data remains challenging but vital for accurate analysis. Automatic ROI detection can significantly reduce the number of pixels to be processed, speed up the analysis, improve accuracy, and reduce dependency on pathologists [Hossain et al., 2023].

The challenge of acquiring image domain annotations for ROI detection is particularly significant due to the knowledge and commitment required of expert pathologists. Pathologists often identify regions in whole slide images with diagnostic relevance rather than examining the entire slide, with a positive correlation between the time spent on these critical image regions and diagnostic accuracy [Ghezloo et al., 2025]. This selective viewing behavior highlights the importance of developing automated systems that can replicate pathologists' diagnostic patterns.

Recent advances in deep learning techniques have offered improvements in computer-aided diagnosis systems for ROI detection. Innovative approaches include generating heatmaps to represent pathologists' viewing patterns during diagnosis and using these to guide deep learning architectures during training [Ghezloo et al., 2025]. Such systems can integrate pathologists' domain expertise to enhance region of interest detection without needing individual case annotations, potentially surpassing traditional approaches based on color and texture image characteristics.

The clinical application of automated ROI detection has been demonstrated in specific diagnostic contexts, such as Human Epidermal Growth Factor Receptor 2 (HER2) grading for breast cancer patients. Existing HER2 grading relies on manual ROI selection, which is tedious, time-consuming, and suffers from inter-observer and intra-observer variability [Hossain et al., 2023]. Studies have found that HER2 grade changes with ROI selection, emphasizing the critical importance of accurate and consistent ROI detection methods. Advanced approaches using Vision Transformer architectures have shown promising results, achieving high accuracy rates for ROI detection while significantly reducing processing time and improving diagnostic agreement with clinical scores [Hossain et al., 2023].

These automated ROI detection methods not only enhance diagnostic accuracy and efficiency but also streamline annotation processes and aid in the training of new pathologists. By incorporating pathologists' expertise through heatmap generation and advanced deep learning architectures, these systems demonstrate the potential to revolutionize digital pathology workflows8while maintaining the quality and reliability of diagnostic assessments [Ghezloo et al., 2025] (Figure 2-4).

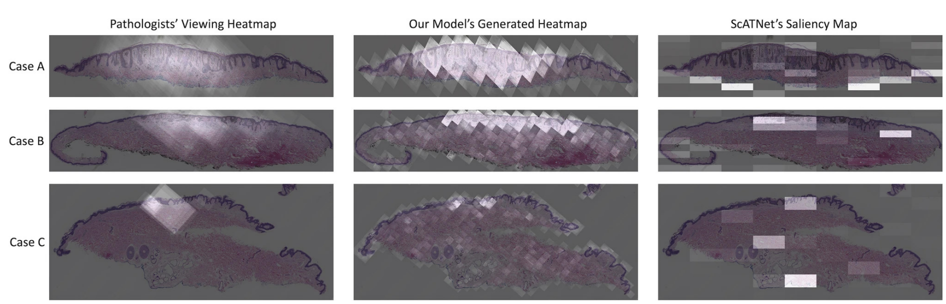


Figure **2-4** ROI heatmaps measured experimentally from pathologists view patterns and estimated heatmap from a U-Net model from [Ghezloo et al., 2025].

2.2.3 Staining Techniques

Histopathological analysis relies heavily on various staining techniques to enhance the visibility of different tissue components and cellular structures. The choice of staining method depends on the specific diagnostic requirements and the type of tissue being examined.

Hematoxylin and Eosin (H&E)

Hematoxylin and Eosin (H&E) staining is the most widely used technique in histopathology, particularly in breast cancer diagnosis [Pan et al., 2021]. This staining method provides essential visualization through two components: hematoxylin, which stains cell nuclei blue/purple to highlight nuclear morphology, and eosin, which stains cytoplasm and extracellular matrix pink/red to reveal tissue architecture.

The popularity of H&E staining in breast cancer histopathology stems from its ability to clearly visualize tumor architecture and growth patterns, distinguish between different types of breast cancer, identify important diagnostic features like nuclear pleomorphism, and assess tumor grade and stage. Beyond breast cancer, H&E staining finds extensive application across various medical specialties including general pathology, dermatology, gastroenterology, neurology, and oncology.

Special Stains

In addition to H&E, various special stains are used for specific diagnostic purposes. Immunohistochemistry (IHC) uses antibodies to detect specific proteins, playing a crucial role in subtyping breast cancers. Key IHC stains include Estrogen Receptor (ER) staining for detecting estrogen receptors, Progesterone Receptor (PGR) staining for assessing progesterone receptor status, Human Epidermal Growth

Factor Receptor 2 (HER2) staining for evaluating HER2 protein expression, and Ki67 staining for measuring cellular proliferation rates. These markers are particularly crucial in breast cancer diagnosis and treatment planning, as they help determine the molecular subtype of the cancer and guide personalized therapeutic approaches. Other specialized stains include Periodic Acid-Schiff (PAS) for highlighting carbohydrates and basement membranes, Masson's Trichrome for distinguishing between collagen and muscle fibers, and silver stains for detecting microorganisms and nerve fibers. These specialized staining techniques complement H&E by providing additional diagnostic information that is crucial for accurate diagnosis and treatment planning [Weitz et al., 2023]. Examples of these staining techniques are shown in Figure 2-8.

2.3 Deep Learning Fundamentals

Deep Learning has emerged as a powerful subset of Machine Learning, revolutionizing the field of artificial intelligence. Its roots can be traced back to the early development of artificial neural networks in the 1940s and 1950s, with significant milestones including the perceptron in 1958 and the backpropagation algorithm in the 1980s. However, it wasn't until the early 21st century, with the advent of more powerful computational resources and the availability of large datasets, that deep learning truly began to flourish [Werbos, 1990].

2.3.1 Learning Paradigms

Machine Learning systems can be categorized into three main learning paradigms. The most common approach is supervised learning, where models learn from labeled data by mapping inputs to known outputs. This paradigm requires a large amount of labeled training data, which can be expensive and time-consuming to acquire. Semi-supervised learning offers a hybrid approach that leverages both labeled and unlabeled data [Alloghani et al., 2020], proving particularly useful when labeled data is scarce but unlabeled data is abundant. Finally, unsupervised learning enables models to discover patterns and structures from unlabeled data without explicit guidance [Alloghani et al., 2020], making it valuable for tasks like clustering and dimensionality reduction. Being deep learning a subset of Machine Learning, it is based on the same paradigms. An illustration of the three main learning paradigms is shown in Figure 2-5.

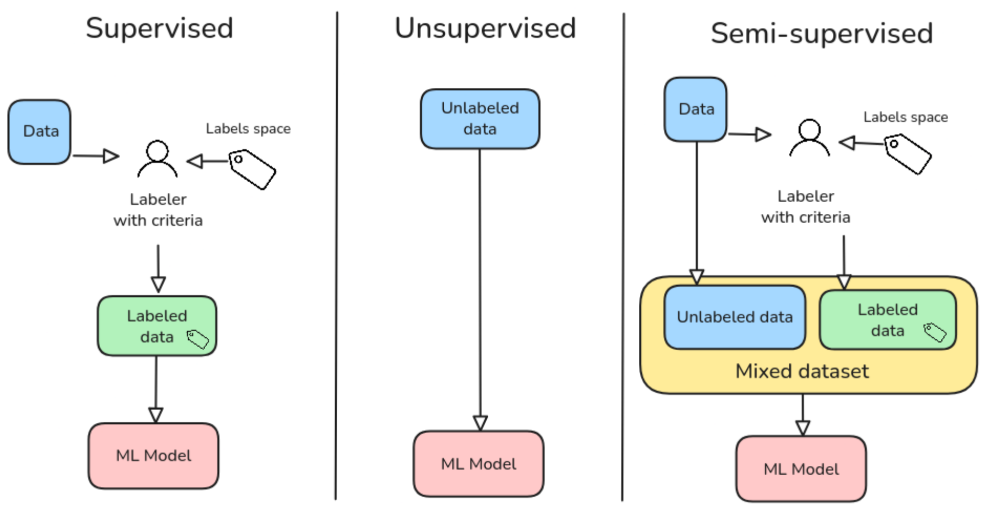


Figure **2-5** Common learning paradigms.

2.3.2 Architecture and Training

Deep Learning architectures are characterized by their layered structure, where each layer progressively extracts and transforms features from the input data [Taye, 2023]. The early layers typically focus on low-level feature extraction, such as edges, textures, and basic patterns in the case of image processing [Taye, 2023]. As information flows through the network, middle layers combine these basic features into more complex representations. The final layers perform high-level reasoning and make the ultimate predictions or classifications.

The training process commonly employs gradient descent as its primary optimization strategy, which iteratively adjusts the model's parameters to minimize a loss function. However, other optimization algorithms like Adam, RMSprop, and evolutionary strategies [Soydaner, 2020] [Bingham et al., 2020], can also be effective for training deep networks. The loss function serves as a crucial component of the learning process in most optimization techniques, quantifying how well the model's predictions match the actual targets. By providing a measure of the model's performance, the loss function guides the chosen optimization process, enabling the network to learn meaningful patterns from the training data.

2.3.3 Challenges and Solutions

Despite their power, deep learning systems face several significant challenges. One of the most prominent issues is overfitting, where models may memorize training data instead of learning generalizable patterns [Roelofs et al., 2019]. This challenge is typically addressed through various regularization techniques such as dropout, L1/L2 regularization, and early stopping [Roelofs et al., 2019]. Another critical challenge is the substantial data requirements; deep learning models often need massive amounts of training data to achieve good performance [Roelofs et al., 2019], which can be a limiting factor in many applications. Additionally, the complex, layered nature of deep learning models makes them difficult to interpret, often referred to as "black boxes". This lack of transparency can be particularly problematic in critical applications where understanding the decision-making process is essential.

2.3.4 Deep Learning Frameworks

The development of powerful open-source frameworks has significantly accelerated deep learning research and applications. TensorFlow, developed by

Google, provides a comprehensive ecosystem for building and deploying Machine Learning models [Abadi et al., 2016]. PyTorch, created by Facebook's AI Research lab, offers dynamic computation graphs and has become particularly popular in research settings [Paszke et al., 2019]. Caffe, known for its speed and modularity, is widely used in computer vision applications [Jia et al., 2014].

These frameworks have democratized deep learning by providing efficient implementations of common operations, automatic differentiation for gradient computation, and GPU acceleration for faster training. They also offer pre-trained models and transfer learning capabilities, along with active communities for support and knowledge sharing. The combination of these frameworks with modern hardware has enabled researchers and practitioners to develop increasingly sophisticated models, pushing the boundaries of what's possible in artificial intelligence. TensorFlow and PyTorch have emerged as the two most prominent frameworks in the deep learning landscape, as shown in Figure 2-6, which presents data from Google Trends over the last five years (as of April 2025).

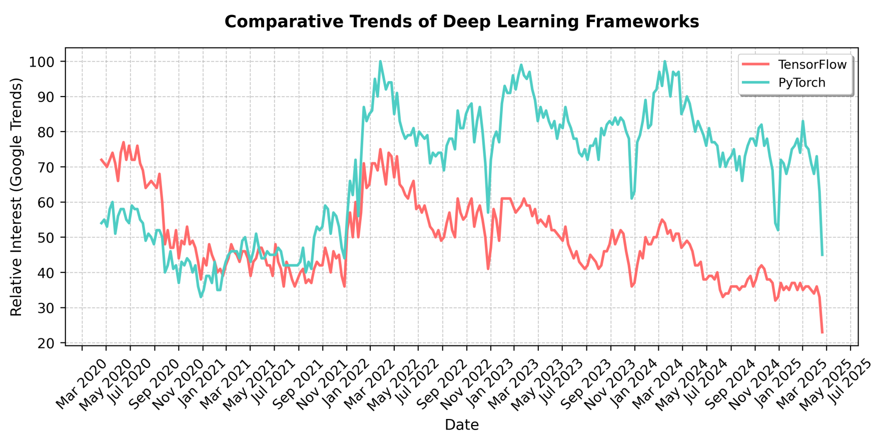


Figure **2-6** Comparative Trends of the top two most popular Deep Learning Frameworks.

2.4 Datasets and Data Sources

Throughout the development of this work, multiple datasets were used for evaluation of ISS models. The common elements of all these datasets are that they contain RGB images and are crowdsourced with multiple labelers, where not necessarily all labeler label all images.

As it has been mentioned in Chapter 1, the main goal of this work is mainly focused on crowdsourced histopathology images semantic segmentation, however, these datasets present the following challenges:

- Distribution of segmentation labels is not uniform across the image, since some tissues and structures are more common than others.
- Visualization of performance of the models in debug time (like per epoch analysis) is not simple for non experts in the subject, which makes it hard to evaluate whether the model is overfitting or not at a glance.

For these reasons, multiple datasets were created this work in the pursuit of an initial evaluation of performance of the models against more traditional and familiar images before the focus on histopathology images. Once a satisfactory performance was achieved, the focus was shifted to histopathology images and further tunings on the models were performed if needed.

In any case, both the emulated noisy annotations datasets and the histopathology datasets somehow contained ground truth aggregation, either from the original source (in the case of emulated noisy annotations), the expert annotation (if available) or from the aggregation of multiple labelers [1].

2.4.1 Oxford-IIIT Pet Dataset

Using the techniques described later in Section 5.2.1, the Oxford-IIIT Pet Dataset [Parkhi et al., 2012] was used to create a dataset with emulated noisy annotations. This dataset is officially available in the 'tensorflow_datasets' package, making it convenient to use in experiments. It is composed of 7349 images of pets, being 2371 of these images cats, and the rest 4978, dogs. Breed labels are included as well as segmentation masks for the classes: 'background', 'border', and 'pet'. Figure 2-7 shows an example of the annotations in the original dataset.

[1] STAPLE in the case of histopathology datasets with no expert annotations available

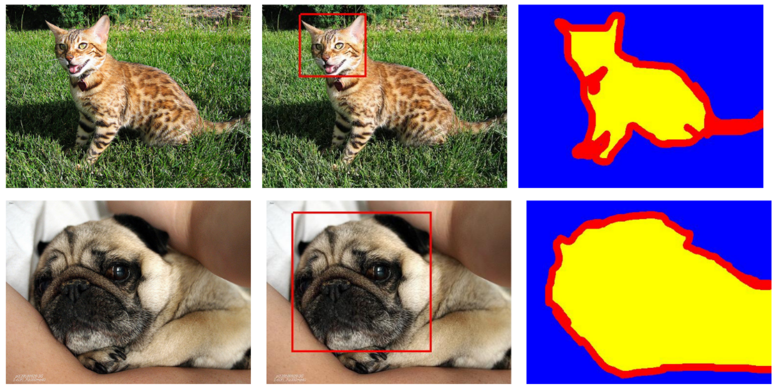


Figure **2-7** Annotations in the Oxford-IIIT Pet data. From left to right: pet image, head bounding box, and trimap segmentation (blue: background region; red: ambiguous region; yellow: foreground region).

2.4.2 Real histopathology datasets

Multi-Stain Breast Cancer Histological Dataset

The Multi-Stain Breast Cancer Histological Dataset [Weitz et al., 2023] represents one of the largest publicly available collections of whole slide images (WSIs) from surgical resection specimens of primary breast cancer patients. This dataset is particularly valuable for our work because it contains matched pairs of H&E and IHC-stained tissue sections from the same tumor, with a total of 4,212 WSIs from 1,153 patients. The IHC stains include important biomarkers such as ER (Estrogen Receptor), PGR (Progesterone Receptor), HER2 (Human Epidermal Growth Factor Receptor 2), and KI67 (Ki67 Proliferation Marker), which are routinely used in breast cancer diagnosis and treatment planning (more on staining techniques in Section 2.2.3).

The dataset's relevance to our work stems from several key aspects. The matched H&E and IHC stains allows for studying the consistency of segmentation across different staining modalities, which is crucial for understanding how different visualization methods affect annotation quality. With 1,153 patients, the dataset provides a robust foundation for training and evaluating segmentation models in a real-world clinical setting. The inclusion of routine diagnostic cases makes the dataset representative of actual clinical practice, where variations in staining quality and tissue preparation are common. Furthermore, the multiple biomarker stains (ER, PGR, HER2, KI67) enable to study how different tissue characteristics affect segmentation performance and annotator agreement.

This dataset serves as an ideal testbed for our crowdsourced segmentation approach. It allows us to evaluate how different staining modalities affect annotator performance and agreement, while also providing insights into the relationship between tissue characteristics and segmentation difficulty. The dataset's comprehensive nature enables validation of our models' performance across different biomarker expressions and assessment of the generalizability of our approach to real-world clinical data.

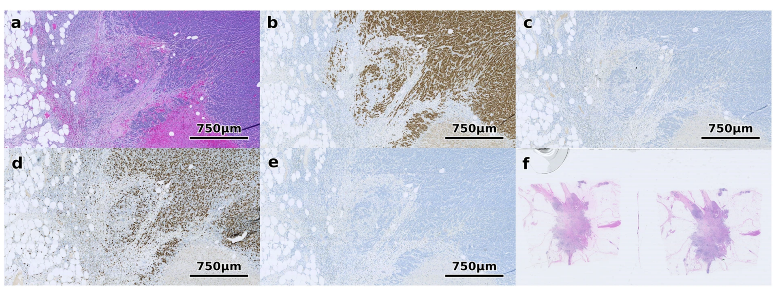


Figure **2-8** Different staining techniques obtained from multi-stain breast cancer dataset [Weitz et al., 2023]. (a) shows H&E, (b) ER, (c) HER2, (d) Ki67 and (e) PGR. (f) shows an example of a WSI that was excluded since it contains multiple tissue sections.

Structured Crowdsourcing Dataset for Histology Images

The dataset presented by [Amgad et al., 2019] is particularly relevant to our work as it represents one of the first systematic studies of crowdsourced annotations in histopathology. The authors recruited 25 participants with varying levels of expertise (from senior pathologists to medical students) to delineate tissue regions in 151 breast cancer slides using the Digital Slide Archive platform, resulting in over 20,000 annotated tissue regions.

Key aspects of this dataset make it valuable for our work. The systematic evaluation of inter-participant discordance revealed varying levels of agreement across different tissue classes, with low discordance for tumor and stroma, and higher discordance for more subjectively defined or rare tissue classes. The inclusion of feedback from senior participants helped in curating high-quality annotations, demonstrating that fully convolutional networks trained on these crowdsourced annotations can achieve high accuracy (mean AUC=0.945). The dataset also provides evidence that the scale of annotation data significantly improves image classification accuracy.

This dataset is particularly valuable for our work because it provides crucial insights into how annotator expertise affects segmentation quality. It demonstrates the feasibility of using crowdsourced annotations for training accurate segmentation models, showing that even with varying levels of expertise, aggregated annotations can produce reliable ground truth. The dataset includes a systematic analysis of inter-annotator agreement, which is crucial for understanding the challenges in crowdsourced histopathology segmentation and informing the development of more robust segmentation approaches.

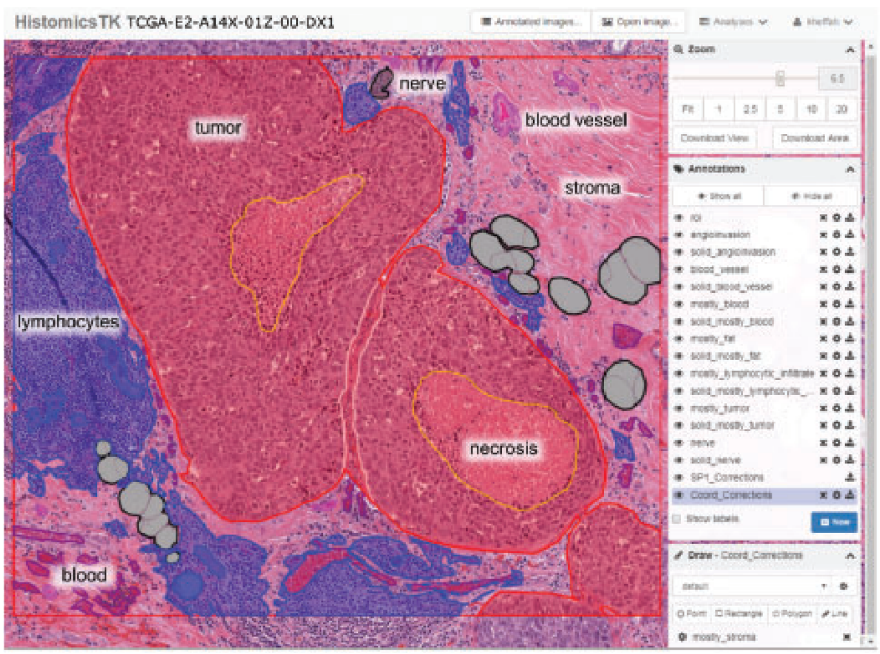


Figure **2-9** Screenshot of the DSA and HistomicsTK web interface while creating the crowdsourced annotations for the dataset presented by [Amgad et al., 2019].

CHAPTER

THREE

CHAINED GAUSSIAN PROCESSES

3.1 Background and Related Methods

In recent years, several approaches have been proposed to handle supervised learning problems in the context of multiple annotators [Yan et al., 2014] [Zhang, 2022]. This section reviews key methods that form the foundation for understanding chained Gaussian processes and their application to multiple annotator scenarios in regression and classification tasks.

3.1.1 Kernel Alignment-Based Annotator Relevance Analysis

The Kernel Alignment-Based Annotator Relevance Analysis (KAAR) method, introduced in [Gil-Gonzalez et al., 2018], addresses the challenge of estimating annotator expertise in scenarios where the ground truth is unavailable. The key innovation of KAAR lies in its use of Centered Kernel Alignment (CKA) to measure the similarity between input features and annotator labels.

Given a dataset with input features $\boldsymbol{X}$ and labels from multiple annotators $\boldsymbol{Y}$, KAAR computes a kernel matrix $\boldsymbol{K_\nu}$ as a convex combination of R basis kernels:

$$\boldsymbol{K_\nu} = \sum_{r=1}^{R} \nu_r \boldsymbol{K}_r, \tag{3-1}$$

where $\boldsymbol{K}_r$ is computed using a kernel function over the input features, and ν_r represents the relevance weight for the r-th annotator. The weights are optimized by maximizing the CKA between $\boldsymbol{K_\nu}$ and a target kernel $\boldsymbol{F}$ computed from the input features:

$$\rho(\boldsymbol{K_\nu}, \boldsymbol{F}) = \frac{\langle \bar{\boldsymbol{K}}_{\boldsymbol{\nu}}, \bar{\boldsymbol{F}} \rangle_{\mathsf{F}}}{||\, \bar{\boldsymbol{K}}_{\boldsymbol{\nu}} \,||_{\mathsf{F}} \,||\, \bar{\boldsymbol{F}} \,||_{\mathsf{F}}}, \tag{3-2}$$

A key advantage of KAAR is its ability to capture dependencies between annotators through the kernel framework, without requiring explicit parametric modeling of these relationships.

3.1.2 Localized Kernel Alignment-Based Annotator Relevance Analysis

In [Gil-Gonzalez et al., 2021a] the authors extend KAAR by introducing locality in the kernel alignment framework. Rather than assuming constant annotator performance across the input space, Localized Kernel Alignment-Based Annotator Relevance Analysis (LKAAR) models the annotator relevance as a function of the input features. The kernel combination in LKAAR takes the form:

$$\boldsymbol{K}_{\boldsymbol{q}} = \sum_{r=1}^{R} \boldsymbol{Q}_r \boldsymbol{K}_r \boldsymbol{Q}_r, \tag{3-3}$$

where $\boldsymbol{Q}_r$ is a diagonal matrix whose elements represent the local relevance of annotator r. These elements are computed as:

$$q_r(\boldsymbol{x}_n) = \begin{cases} \beta_0^{(r)} + \sum_{n'=1}^{N} \beta_{n'}^{(r)} \kappa_\beta(\boldsymbol{x}_n, \boldsymbol{x}_{n'}), & \text{if } n \in N_r \\ 0, & \text{Otherwise} \end{cases} \tag{3-4}$$

This localized approach allows LKAAR to better model inconsistent annotators whose performance varies across different regions of the input space [Gil-Gonzalez et al., 2021a].

3.1.3 Chained Gaussian Processes Model

The CGP model provides a probabilistic framework for modeling multiple outputs through a chain of Gaussian processes [Saul et al., 2016]. In the context of multiple annotators, CGP can be used to model both the ground truth and annotator behavior. The key idea is to model a likelihood function with J parameters as in [Giraldo et al., 2022]:

$$p(\boldsymbol{y}|\boldsymbol{X}, \boldsymbol{\theta}) = \prod_{n=1}^{N} p(y_n|\theta_1(\boldsymbol{x}_n), \ldots, \theta_J(\boldsymbol{x}_n)), \tag{3-5}$$

where each parameter $\theta_j(\boldsymbol{x})$ is modeled using a separate Gaussian process. This framework provides remarkable flexibility in modeling complex relationships within the data [Giraldo et al., 2022]. The use of Gaussian processes enables natural uncertainty quantification through posterior distributions, allowing the model to express varying degrees of confidence in its predictions. The framework inherently handles missing data through its probabilistic nature, and the automatic relevance determination property of GPs helps identify the most important features for prediction [Giraldo et al., 2022].

The CGP framework becomes particularly powerful when integrated with concepts from KAAR and LKAAR. By incorporating kernel alignment principles, the model can make more informed choices about kernel functions, better capturing the underlying structure of the data [Giraldo et al., 2022]. The framework can be extended to include input-dependent lengthscales, allowing for varying degrees of smoothness across the input space [Giraldo et al., 2022]. Furthermore, through multi-output Gaussian Process (GP) formulations, the model can capture complex dependencies capture complex dependencies between different annotators, providing a rich representation of how different experts' opinions relate to each other.

However, implementing CGPs in practice presents several significant challenges. The computational complexity of the model scales cubically with the dataset size, making it potentially prohibitive for large-scale applications [Liu et al., 2020]. The need for approximate inference techniques in non-Gaussian likelihood scenarios can introduce additional complexity and potential approximation errors [Rasmussen and Williams, 2006a]. When dealing with high-dimensional inputs, the model may struggle with the curse of dimensionality, requiring careful feature selection or dimensionality reduction preprocessing [Rasmussen and Williams, 2006a]. Additionally, the selection of appropriate kernel functions remains a critical challenge, as it significantly impacts the model's performance and its ability to capture relevant patterns in the data [Rasmussen and Williams, 2006b].

3.1.4 Regularized Chained Deep Neural Network

In [Gil-Gonzalez et al., 2021b], the authors propose a deep learning approach to multiple annotator learning, inspired by the chained Gaussian processes model. The method models both the ground truth estimation and annotator reliability through a deep neural network architecture. The likelihood function in Regularized Chained Deep Neural Network (RCDNN) takes the form:

$$p(\boldsymbol{Y}|\boldsymbol{\theta}) = \prod_{n=1}^{N} \prod_{r \in R_n} \left(\prod_{k=1}^{K} \zeta_{n,k}^{\delta(y_n^{(r)}-k)} \right)^{\lambda_n^{(r)}} \left(\frac{1}{K} \right)^{1-\lambda_n^{(r)}}, \tag{3-6}$$

where $\lambda_n^{(r)}$ represents the reliability of annotator r for instance n, and $\zeta_{n,k}$ is the estimated probability of class k for instance n.

The RCDNN architecture is built upon a sophisticated multi-layer structure that combines several key innovations in deep learning [Gil-Gonzalez et al., 2021b]. At its core, it employs multiple dense layers with carefully chosen activation functions, each designed to capture different aspects of the annotator-data relationship. To prevent overfitting, a comprehensive regularization strategy is implemented, combining both L1 and L2 norms with dropout mechanisms [Gil-Gonzalez et al., 2021b]. The network architecture splits into specialized branches: one dedicated to estimating the ground truth labels, and another focused on modeling annotator reliability. This dual-branch approach allows the network to simultaneously learn both the true underlying patterns in the data and the individual characteristics of each annotator. Furthermore, the implementation of Monte Carlo dropout during prediction provides a mechanism for uncertainty estimation, offering not just predictions but also confidence measures for those predictions.

3.2 Tooling and Implementation

The implementation of Chained Gaussian Processes for multiple annotator learning requires robust and flexible software tools that can handle both the probabilistic nature of Gaussian processes and the complexity of deep architectures [Saul et al., 2016]. In this work, two key frameworks are leveraged: GPflow and GPflux. These tools provide the foundation for building scalable and efficient implementations of our models while maintaining the theoretical rigor necessary for proper uncertainty quantification.

The choice of these frameworks is motivated by several factors. First, both are built on top of TensorFlow, providing access to automatic differentiation and GPU acceleration capabilities. Second, they offer a modular design that allows for easy extension and modification, which is crucial when implementing novel model architectures. Third, they provide robust implementations of various Gaussian process models and inference methods, allowing us to focus on the specific challenges of multiple annotator learning rather than reimplementing basic GP functionality.

3.2.1 GPflow Overview

GPflow is a Python package for building Gaussian process models in TensorFlow. Developed as a modern and scalable alternative to GPy, it leverages TensorFlow's computational capabilities to provide efficient implementation of Gaussian process models. The framework is designed with four key principles in mind: ease of use, computational efficiency, extensibility, and automatic differentiation. This work was originally published in [de G. Matthews et al., 2016].

At its core, GPflow provides implementations of various Gaussian process models including Gaussian Process Regression (GPR), Gaussian Process Monte Carlo (GPMC), Sparse Gaussian Process Regression (SGPR), and Sparse Gaussian Process Monte Carlo (SGPMC) [de G. Matthews et al., 2016]. These implementations are built on top of TensorFlow's computational graph framework, allowing for automatic differentiation and GPU acceleration where available.

The framework's architecture is modular, with clear separation between model components such as kernels, likelihoods, and optimization routines. This modularity makes it straightforward to extend existing models or create new ones. GPflow includes a comprehensive suite of kernel functions, ranging from the basic RBF (Radial Basis Function) kernel to more sophisticated ones like the Spectral Mixture kernel.

One of GPflow's distinguishing features is its robust handling of model parameters. It provides facilities for parameter transformation (ensuring parameters remain in valid ranges during optimization), prior specification, and flexible optimization strategies. The framework also includes utilities for model evaluation and prediction, with built-in support for uncertainty estimation.

3.2.2 GPflux Framework

GPflux extends GPflow to enable the construction and training of Deep Gaussian Process (DGP) models using modern deep learning practices. Built on top of both GPflow and TensorFlow, GPflux provides a flexible framework for implementing deep probabilistic models that combine the expressiveness of deep neural networks with the uncertainty quantification capabilities of Gaussian processes. This work was originally published in [Dutordoir et al., 2021].

The framework implements Deep Gaussian Processes as a composition of Gaussian process layers, where each layer transforms its inputs through a Gaussian process mapping. This hierarchical structure allows the model to capture complex, non-stationary patterns in data while maintaining uncertainty estimates throughout the entire processing pipeline. GPflux handles this complexity through a carefully designed architecture that separates the concerns of model specification, inference, and prediction.

A key innovation in GPflux is its integration with Keras-style layers, allowing practitioners to combine deterministic neural network layers with Gaussian process layers in a single model. This hybrid approach enables the construction of models that can leverage both the structural assumptions encoded in neural network architectures and the flexibility of Gaussian processes.

For inference, GPflux implements various approximate methods, including stochastic variational inference with inducing points, which makes it possible to train deep Gaussian process models on large datasets. The framework provides built-in support for minibatch training, making it practical to work with datasets that don't fit in memory.

3.3 Experimental Setup

Synthetic datasets were used to evaluate the proposed methods in classification and regression tasks, following very similar procedures to the ones used in [Gil-Gonzalez et al., 2023], which are as follows:

3.3.1 Classification

A fully synthetic data for a 1-D ($P = 1$) multiclass classification problem with $K = 3$ classes is generated. The input feature matrix $\mathbf{X}$ is constructed by randomly sampling $N = 100$ points from a uniform distribution over the interval $[0, 1]$. For each sample n, the true label is determined by computing three target values:

$$\begin{aligned} t_{n,1} &= \sin(2\pi x_n), \\ t_{n,2} &= -\sin(2\pi x_n), \\ t_{n,3} &= -\sin(2\pi(x_n + 0.25)) + 0.5, \end{aligned} \tag{3-7}$$

and assigning the class label as the index $i \in \{1, 2, 3\}$ that maximizes $t_{n,i}$. This construction ensures a nontrivial, nonlinear decision boundary between the classes [Gil-González and Álvarez Meza, 2023].

For evaluation, the test set is generated by extracting 200 equally spaced samples from the interval $[0, 1]$, providing a dense coverage of the input space to assess generalization performance.

Note that at this point, the fully synthetic dataset does not contain real annotator labels. Therefore, it is necessary to simulate annotator labels as corrupted versions of the hidden ground truth. In these experiments, the simulation is designed to account for: 1) dependencies among annotators, and 2) labeler performance that varies as a function of the input features.

To achieve this, an SLFM-based approach (termed SLFM-C) is employed, which generates annotator labels as follows:

1. **Definition of deterministic functions:** Define Q deterministic functions $\hat{\mu}_q : \mathcal{X} \to \mathbb{R}$ and their combination parameters $\hat{w}_{l,r,q} \in \mathbb{R}$, for all $r \in R$, $n \in N$.

2. **Annotator performance computation:** Compute $\hat{f}_{l,r,n} = \sum_{q=1}^{Q} \hat{w}_{l,r,q}\hat{\mu}_q(\hat{x}_n)$, where $\hat{x}_n \in \mathbb{R}$ is the nth component of $\hat{\mathbf{x}} \in \mathbb{R}^N$. Here, $\hat{\mathbf{x}}$ is the 1-D representation of the input features in $\mathbf{X}$, obtained using the t-distributed Stochastic Neighbor Embedding (t-SNE) approach.

3. **Reliability calculation:** Calculate $\hat{\lambda}_n^r = \varsigma(\hat{f}_{l,r,n})$, where $\varsigma(\cdot) \in [0, 1]$ is the inverse exponential sigmoid function.

4. **Label assignment:** The rth annotator's label for the nth sample is given by
$$y_n^r = \begin{cases} y_n, & \text{if } \hat{\lambda}_n^r \geq 0.5 \\ \tilde{y}_n, & \text{if } \hat{\lambda}_n^r < 0.5 \end{cases}$$
where y_n is the true label and $\tilde{y}_n$ is a flipped version of the actual label y_n.

This procedure ensures that the simulated annotator labels exhibit both inter-annotator dependencies and input-dependent reliability.

3.3.2 Regression

A fully synthetic dataset is also generated for a 1-D regression problem, where the ground truth for the nth sample is defined as:

$$y_n = \sin(2\pi x_n)\sin(6\pi x_n) \tag{3-8}$$

Here, the input matrix $\mathbf{X}$ is constructed by randomly sampling 100 points uniformly from the interval $[0, 1]$. The test set is created by extracting equally spaced samples from the same interval, ensuring a clear separation between training and testing data.

To simulate the presence of multiple annotators, the ground truth is corrupted with annotator-specific noise. For each annotator r, the observed label for the nth sample is given by:

$$y_n^r = y_n + \epsilon_n^r$$

where $\epsilon_n^r \sim \mathcal{N}(0, v_n^r)$ is Gaussian noise, and v_n^r represents the error variance for the rth annotator, which may depend on the input features. This setup allows us to model both the bias and the reliability of each annotator, as well as potential dependencies between annotators.

To further increase realism, the error variance v_n^r can be modeled as a function of the input features, capturing scenarios where annotator reliability varies across the input space, thus:

$$\epsilon_n^r \sim \mathcal{N}(0, v_n^r)$$

being v_n^r, the rth annotator's error variance for the nth sample.

Again, for modeling the error as a function of the input features, an SLFM-based approach (termed SLFM-R) is employed to build the noisy labels, as follows:

1. **Definition of deterministic functions:** Define Q deterministic functions $\hat{\mu}_q : \mathcal{X} \to \mathbb{R}$ and their combination parameters $\hat{w}_{l,r,q} \in \mathbb{R}$, for all $r \in R, n \in N$.
2. **Annotator performance computation:** Compute $\hat{f}_{l,r,n} = \sum_{q=1}^{Q} \hat{w}_{l,r,q}\hat{\mu}_q(\hat{x}_n)$, where $\hat{x}_n \in \mathbb{R}$ is the nth component of $\hat{\mathbf{x}} \in \mathbb{R}^N$. Here, $\hat{\mathbf{x}}$ is the 1-D representation of the input features in $\mathbf{X}$, obtained using the t-distributed Stochastic Neighbor Embedding (t-SNE) approach.
3. **Compute variance :** Finally, determine $v_n^r = \exp(\hat{f}_{l,r,n})$.

3.4 Results

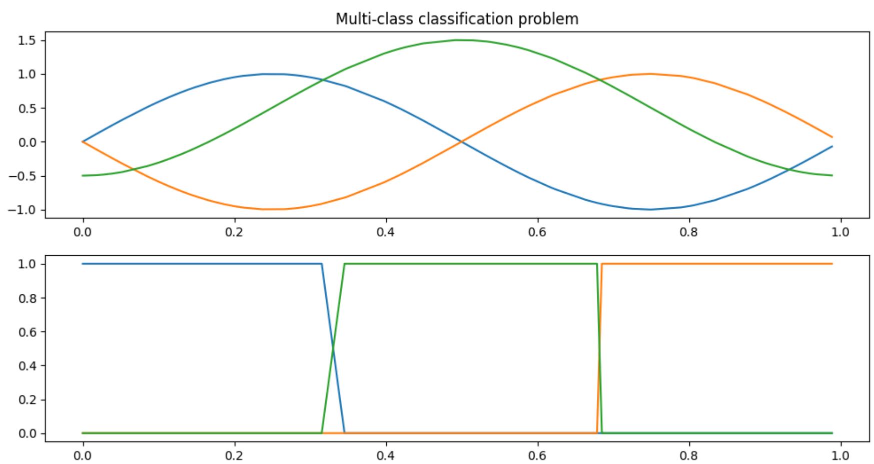
The built synthetic datasets are used to train the CCGPMA model. Emulated classification labels following the true label determining function from 3-7 can be seen in Figure **3-1**. On the other hand, the ground truth and noisy labels emulation for annotators across the feature space for regression task can be seen in Figure **3-3**.

Using both GPflux and GPflow for the regression and classifications tasks, leads to the following combinations experiments: GPflux-CCGPMA Regression, GPflow-CCGPMA Regression, GPflux-CCGPMA Binary Classification, GPflow-CCGPMA Binary Classification, GPflow-CCGPMA Multiclass Classification, GPflux-CCGPMA Multiclass Classification. All of the experiments converged successfully with almost identical results and close to no difference in performance.

3.5 Conclusion

3.5.1 Comparative Analysis

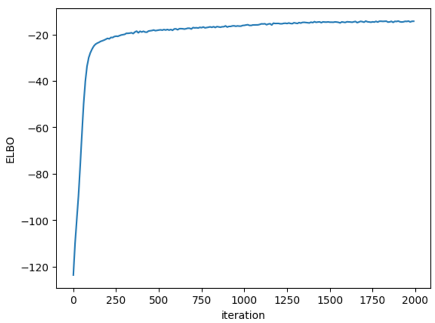
The evolution of multiple annotator learning methods represents a fascinating progression in how we approach the challenge of learning from diverse expert



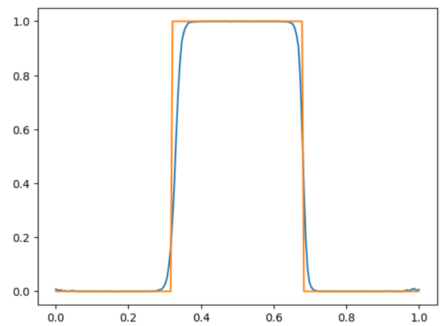
(a) Ground truth labels across the feature space

(b) Noisy emulations for annotators across the feature space for classification task

Figure **3-1** Synthetic dataset overview for classification task



(a) ELBO convergence



(b) Trained CCGPMA predictions on the test set

Figure **3-2** Classification results for the CCGPMA model using GPflux.

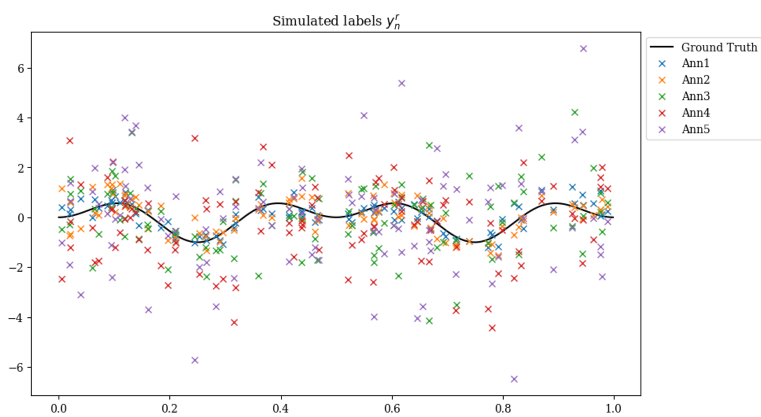
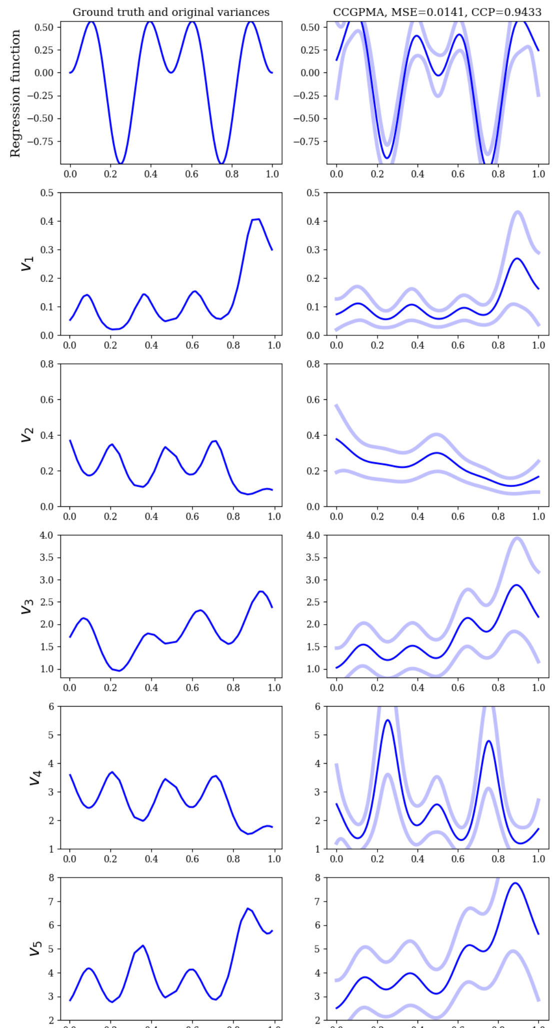


Figure **3-3** Ground truth and noisy labels emulation for annotators across the feature space for regression task.



(a) Trained CCGPMA predictions on the test set, for inference of both the hidden ground truth and the annotators' reliabilities across the feature space.

Figure **3-4** Regression results for the CCGPMA model using GPflux.

opinions. KAAR established a foundation with its non-parametric approach through kernel alignment, though it operates under the assumption of constant annotator performance across the input space. LKAAR advanced this concept by introducing input-dependent annotator behavior, recognizing that expert performance often varies across different types of inputs. RCDNN brought the power and flexibility of deep learning to the problem, offering excellent scalability but requiring careful attention to regularization and architecture design. CCGPMA represents perhaps the most theoretically elegant approach, providing a principled probabilistic framework, though it faces significant computational challenges in practical applications.

The selection of an appropriate method for a given application requires careful consideration of several factors. The size and dimensionality of the dataset play a crucial role, as some methods scale better than others with increasing data volume. The need for uncertainty quantification may favor probabilistic approaches like CGP, while computational constraints might push towards more efficient methods like KAAR. The presence of input-dependent annotator behavior could make LKAAR or RCDNN more appropriate choices. Additionally, when interpretability is a key requirement, as in many medical applications, methods like KAAR and LKAAR might be preferred over the more complex Deep Learning approaches. Table **3-1** provides a comprehensive comparison of the key characteristics and capabilities of each method discussed.

Table **3-1** Comparison of Multiple Annotator Learning Methods

Characteristic	**KAAR** [Gil-Gonzalez et al., 2018]	**LKAAR** [Gil-Gonzalez et al., 2021a]	**RCDNN** [Gil-Gonzalez et al., 2021b]	**CGP** [Gil-González and Álvarez Meza, 2023]
Input-dependent annotator behavior	No	Yes	Yes	Yes
Annotator dependencies modeling	Yes	Yes	Yes	Yes
Uncertainty quantification	No	No	Partial	Yes
Computational scalability	High	Medium	High	Low
Missing data handling	Yes	Yes	Yes	Yes
Non-linear relationships	Yes	Yes	Yes	Yes
Training complexity	Low	Medium	High	High
Hyperparameter sensitivity	Low	Medium	High	Medium
Interpretability	High	Medium	Low	Medium
Memory requirements	Low	Medium	High	High

Note: Partial uncertainty quantification in RCDNN refers to the use of Monte Carlo dropout for approximate uncertainty estimation.

3.5.2 Implementation Considerations

When implementing Chained Gaussian Processes for multiple annotator learning using these tools, several key considerations must be addressed. First, the model architecture must be carefully designed to capture both the individual annotator characteristics and their interdependencies [Gil-González and Álvarez Meza, 2023]. This requires extending the basic GP layers provided by GPflux to handle multiple output streams - one for each annotator's reliability model.

Second, the inference procedure must be adapted to handle the unique challenges of multiple annotator data, such as missing labels and varying levels of annotator expertise across different regions of the input space. The variational inference capabilities of GPflux provide a solid foundation for this, but careful attention must be paid to the design of the variational approximation to ensure it captures the relevant uncertainty in both the ground truth and annotator reliability estimates.

Third, kernel selection and design presents a critical challenge in implementing the CGP model. The choice of kernel functions significantly impacts the model's ability to capture the underlying structure of the data and the relationships between annotators [Rasmussen and Williams, 2006b]. Several considerations must be taken into account:

- **Kernel Complexity**: The model requires multiple kernel functions to capture different aspects of the data - one for the ground truth and additional kernels for each annotator's reliability model. This multi-kernel setup increases the complexity of kernel selection and optimization.

- **Input-Dependent Behavior**: Since annotator performance often varies across the input space, the kernel functions must be flexible enough to capture this non-stationary behavior. This may require using composite kernels or kernels with input-dependent lengthscales.

- **Inter-Annotator Dependencies**: The kernel design must account for potential correlations between different annotators' behaviors. This can be achieved through careful design of the kernel structure in the SLFM framework, where the combination of basis kernels must be chosen to effectively model these dependencies.

- **Computational Efficiency**: The choice of kernel functions also impacts computational performance. While more expressive kernels may better capture complex patterns, they often come with increased computational cost. This trade-off must be carefully considered, especially when dealing with large datasets.

Finally, computational efficiency becomes a critical concern when scaling to large datasets with multiple annotators. The minibatch training capabilities of both frameworks help address this, but additional considerations such as appropriate inducing point selection and batch size tuning become important for achieving good performance in practice.

3.5.3 Final Remarks

In this chapter, we have presented a comprehensive overview of the Chained Gaussian Processes model and its application to multiple annotator learning. We have discussed the theoretical underpinnings of the model, its implementation using GPflow and GPflux, and its evaluation on synthetic datasets. It was shown in [Gil-Gonzalez et al., 2023] that CCGPMA outperforms other method in state of the art methods in terms of accuracy and uncertainty quantification (see 1.3.1).

While Gaussian Processes have demonstrated remarkable success in regression and classification tasks, their application to image segmentation presents significant computational challenges [Rasmussen and Williams, 2006b]. The cubic computational complexity of GPs ($\mathcal{O}(N^3)$), which scales with the number of data points, becomes particularly problematic when dealing with high-resolution images where each pixel can be considered a data point [Liu et al., 2020]. This limitation makes traditional GP approaches impractical for large-scale image segmentation tasks, where the number of pixels can easily reach into the millions. Alternative approaches, such as deep learning-based segmentation methods, often provide more computationally efficient solutions for these high-dimensional problems, though at the cost of uncertainty quantification capabilities that GPs naturally provide [Kendall and Gal, 2017] [Wilson et al., 2016].

CHAPTER

FOUR

TRUNCATED GENERALIZED CROSS ENTROPY FOR SEGMENTATION

4.1 Loss Functions for Multiple Annotators

As mentioned in Section 2.3.2, a loss function is a key element for defining the objective function of a deep learning model. The categorical cross-entropy loss is a common loss function for classification tasks. However, in the case of multiple annotators, the categorical cross-entropy loss is not able to handle the varying reliability of the annotators. In this section, we will propose a loss function that is able to handle multiple annotators' segmentation masks while accounting for their varying reliability across different regions of the image.

4.1.1 Generalized Cross Entropy

The Generalized Cross Entropy (GCE) loss function was first introduced by [Zhang and Sabuncu, 2018] as a robust alternative to the standard cross-entropy loss, particularly effective in handling noisy labels. Let us first consider the Cross Entropy (CE) and Mean Absolute Error (MAE) loss functions:

$$MAE(\mathbf{y}, f(\mathbf{x})) = \|\mathbf{y} - f(\mathbf{x})\|_1 \tag{4-1}$$

$$CE(\mathbf{y}, f(\mathbf{x})) = \sum_{k=1}^{K} y_k \log(f_k(\mathbf{x})) \tag{4-2}$$

where $y_k \in \mathbf{y}$, $f_k(\mathbf{x}) \in f(\mathbf{x})$, and $\|\cdot\|_1$ stands for the l_1-norm. Of note, $\mathbf{1}^\top \mathbf{y} = \mathbf{1}^\top f(\mathbf{x}) = 1$, $\mathbf{1} \in \{1\}^K$ being an all-ones vector. In addition, the MAE loss can be rewritten for softmax outputs, yielding:

$$MAE(\mathbf{y}, f(\mathbf{x})) = 2(1 - \mathbf{1}^\top(\mathbf{y} \odot f(\mathbf{x}))) \tag{4-3}$$

where $\odot$ stands for the element-wise product.

The CE loss function exhibits several distinct characteristics that make it particularly sensitive to noisy labels. It is unbounded from above and heavily penalizes confident but wrong predictions, which can lead to instability in the presence of label noise. In contrast, the MAE loss offers a more robust alternative with its bounded nature and symmetric properties in softmax-based representations. While the MAE loss is more resilient to noisy labels, it tends to train more slowly due to its equal weighting of all mistakes regardless of confidence.

The GCE loss function, as defined by the authors in [Zhang and Sabuncu, 2018], provides a flexible framework that bridges these two approaches:

$$GCE(\mathbf{y}, f(\mathbf{x})) = 2\frac{1-(\mathbf{1}^\top(\mathbf{y}\odot f(\mathbf{x})))^q}{q}, \tag{4-4}$$

with $q \in (0,1]$. Remarkably, the limiting case for $q \to 0$ in GCE is equivalent to the CE expression, and when $q = 1$, it equals the MAE loss. In addition, the GCE holds the following gradient with regard to θ:

$$\frac{\partial GCE(\mathbf{y}, f(\mathbf{x};\theta)|k)}{\partial \theta} = -f_k(\mathbf{x};\theta)^{q-1}\nabla_\theta f_k(\mathbf{x};\theta). \tag{4-5}$$

The GCE loss combines the best of both worlds, offering robustness to label noise while maintaining the convexity property necessary for optimization. The truncation parameter q provides a mechanism to control the sensitivity to outliers, allowing for fine-tuning of the loss function's behavior based on the specific characteristics of the dataset.

4.1.2 Extension to Multiple Annotators

In the context of multiple annotators, we need to consider the varying reliability of each annotator across different regions of the image. Let's consider a k-class multiple annotators segmentation problem with the following data representation:

$$\mathbf{X} \in \mathbb{R}^{W\times H}, \{\mathbf{Y}_r \in \{0,1\}^{W\times H\times K}\}_{r=1}^{R}; \quad \hat{\mathbf{Y}} \in [0,1]^{W\times H\times K} = f(\mathbf{X}) \tag{4-6}$$

where the segmentation mask function maps the input to output as:

$$f: \mathbb{R}^{W\times H} \to [0,1]^{W\times H\times K} \tag{4-7}$$

The segmentation masks $\mathbf{Y}_r$ satisfy the following condition for being a softmax-like representation:

$$\mathbf{Y}_r[w,h,:]\mathbf{1}_k^\top = 1; \quad w \in W, h \in H \tag{4-8}$$

4.1.3 Reliability Maps and Truncated GCE

The key innovation in our approach is the introduction of reliability maps Λ_r for each annotator:

$$\left\{\Lambda_r(\mathbf{X};\theta) \in [0,1]^{W\times H}\right\}_{r=1}^{R} \tag{4-9}$$

These reliability maps serve as a mechanism to estimate the confidence of each annotator at every spatial location (w,h) in the image. By learning these maps jointly with the segmentation model, the network gains the ability to adapt to the varying levels of expertise across different regions of the image. This approach allows for dynamic weighting of each annotator's contribution based on their estimated reliability in specific areas, effectively handling cases where annotators might demonstrate different levels of expertise in different parts of the image.

The proposed Truncated Generalized Cross Entropy for Semantic Segmentation (TGCE$_{SS}$) combines the robustness of GCE with the flexibility of reliability maps. Let us first define the key components:

$$\mathcal{L}_{GCE}(\mathbf{Y}_r, f(\mathbf{X};\theta)) = \mathbb{E}_k\left\{\mathbf{Y}_r \circ \left(\frac{\mathbf{1}_{W\times H\times K} - f(\mathbf{X};\theta)^{\circ q}}{q}\right)\right\} \tag{4-10}$$

$$\mathcal{L}_{prior} = \frac{\mathbf{1}_{W\times H} - \left(\frac{1}{K}\mathbf{1}_{W\times H}\right)^{\circ q}}{q} \tag{4-11}$$

where $\mathcal{L}_{GCE}$ represents the GCE loss component and $\mathcal{L}_{prior}$ represents the uniform prior component. The complete TGCE$_{SS}$ loss function is then defined as:

$$\begin{aligned} TGCE_{SS}(\mathbf{Y}_r, f(\mathbf{X};\theta)|_r(\mathbf{X};\theta)) = \mathbb{E}_{r,w,h}\Bigg\{ \\ \Lambda_r(\mathbf{X};\theta) \circ \mathcal{L}_{GCE} + (\mathbf{1}_{W\times H} - \Lambda_r(\mathbf{X};\theta)) \circ \mathcal{L}_{prior}; \\ w \in W, h \in H; r \in R\Bigg\} \end{aligned} \tag{4-12}$$

where $q \in (0, 1)$ controls the MAE or CE level in the same way as in the GCE loss function. The loss function consists of two main components:

- The first term weighted by Λ_r represents the GCE loss for regions where the annotator is considered reliable
- The second term weighted by $(1 - \Lambda_r)$ provides a uniform prior for regions where the annotator is considered unreliable

A summary of the loss function mechanism is shown in Figure 4-1.

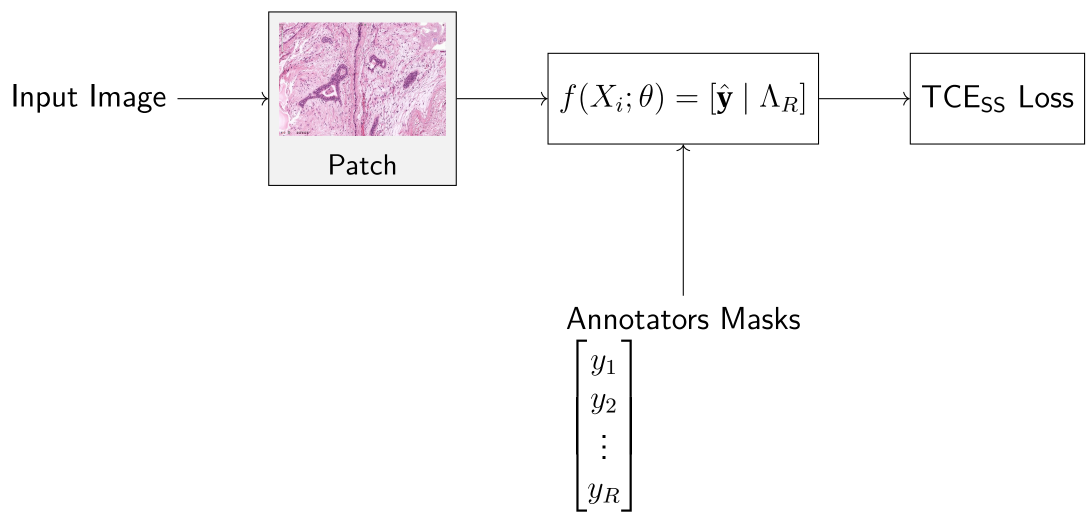


Figure **4-1** Working mechanism of the proposed Truncated Generalized Cross Entropy for Semantic Segmentation (TGCE$_{SS}$) loss function. The loss combines reliability maps Λ_r with the model's predictions $f(\mathbf{X}; \theta)$ to compute a weighted loss that accounts for annotator reliability across different image regions.

4.2 Chapter overview

This chapter presents a novel approach to handle multiple annotators in semantic segmentation tasks through the introduction of the Truncated Generalized Cross Entropy for Semantic Segmentation (TGCE$_{SS}$) loss function. The key contributions and novelties of this work can be summarized as follows:

- **Reliability-aware Learning**: Unlike traditional approaches that treat all annotators equally, the proposed method introduces reliability maps (Λ_r) that dynamically estimate the confidence of each annotator at every spatial location in the image. This allows the model to adapt to varying levels of expertise across different regions.
- **Robust Loss Function**: The proposed TGCE$_{SS}$ combines the benefits of both Cross Entropy and Mean Absolute Error losses through the Generalized Cross Entropy framework. The truncation parameter q provides a flexible mechanism to control the sensitivity to noisy labels while maintaining the convexity property necessary for optimization.
- **Unified Framework**: The approach provides a unified framework that handles both reliable and unreliable regions through a weighted combination of GCE loss and uniform prior. This is particularly important in medical imaging where annotator expertise can vary significantly across different anatomical structures.

The main novelty of this approach lies in its ability to simultaneously learn the segmentation task while estimating annotator reliability in a spatially-aware manner. This represents a significant advancement over existing methods that either:

- Treat all annotators equally without considering their varying expertise
- Use global reliability scores that don't account for spatial variations in annotator performance
- Rely on fixed, pre-defined reliability measures rather than learning them from data

The proposed TGCE$_{SS}$ loss function represents a step forward in handling the complex nature of multiple annotator scenarios in semantic segmentation, particularly in medical imaging applications where expert knowledge can vary significantly across different regions of interest. This work provides a foundation for more sophisticated approaches to handling multiple annotators in deep learning-based segmentation tasks.

CHAPTER

FIVE

CHAINED DEEP LEARNING AND TGCE$_{SS}$ LOSS FOR IMAGE SEGMENTATION

As mentioned in Chapter 1, U-shaped Convolutional Neural Networkss (CNNs) have been proven a good solution for Semantic segmentation (SS) tasks in medical images, due to their ability to capture both global and local information with small datasets.

The proposed model architecture combines the strengths of U-Net with a ResNet-34 backbone, specifically designed to work with the TGCE$_{SS}$ loss function by providing a multi-output network which focuses on two tasks: **Multiclass Semantic Segmentation** and **Reliability Estimation**. The architecture is illustrated in Figure **5-2**.

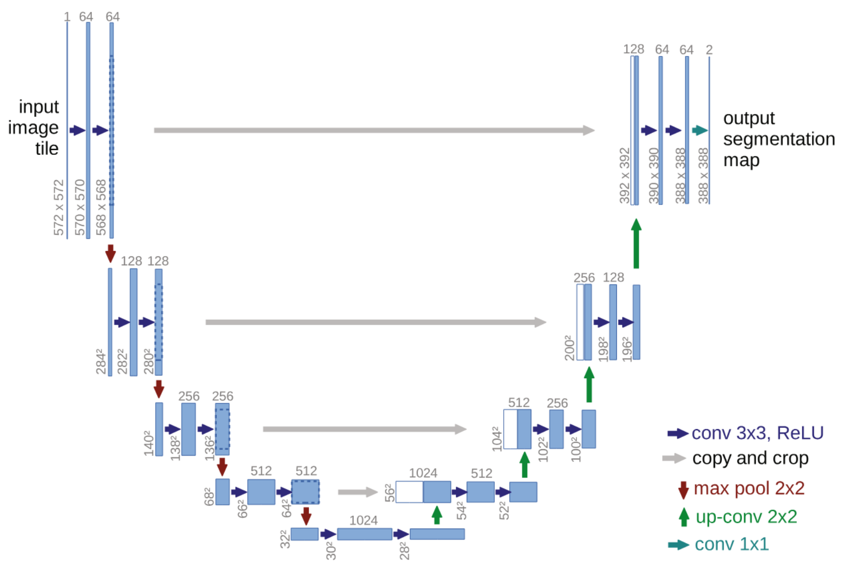


Figure **5-1** Original U-NET architecture from [**?**].

5.1 Backbone Architecture

The model employs a pre-trained ResNet-34 as its encoder backbone, leveraging its deep residual learning framework for efficient feature extraction. The choice of ResNet-34 provides several key advantages: efficient feature extraction through residual connections, pre-trained weights that capture rich visual representations, and stable gradient flow during training. The ResNet-34 backbone is modified to serve as the encoder in our UNET architecture by removing the final fully connected layer and utilizing the feature maps from different stages of the network for skip connections.

5.1.1 U-Net Architecture

The U-Net architecture follows a traditional encoder-decoder structure with skip connections, where the encoder path implements the ResNet-34 structure. The

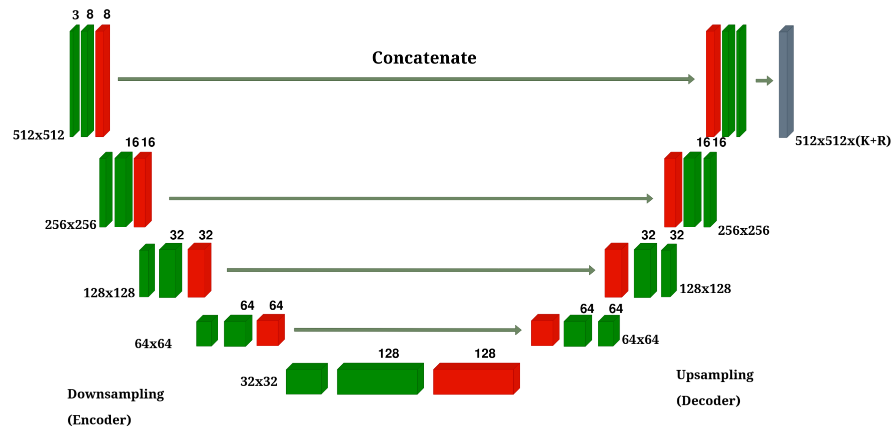


Figure **5-2** Overview of the proposed model architecture.

decoder path employs transposed convolutions for upsampling, creating a symmetrical architecture that effectively captures both high-level and low-level features. The architecture incorporates four downsampling stages in the encoder, corresponding to the ResNet-34 blocks, and four upsampling stages in the decoder. These stages are connected through skip connections that bridge corresponding encoder and decoder stages, allowing the network to preserve fine-grained details. Each convolution operation is followed by batch normalization and ReLU activation to ensure stable training and effective feature learning.

This U-Net architecture is expected to be coupled with a reliability branch which leads to a multi-output network. Three different approaches are considered for this branch: **scalar**, **features**, and **pixel**.

5.1.2 Scalar Reliability Estimation

The scalar reliability estimation branch is the simplest one, in which the reliability branch is a parallel branch to the segmentation branch, which leads to a multi-output network. The final output consists of R reliability maps Λ_r, one for each annotator, with values constrained to the $[0, 1]$ range through a sigmoid activation function. This design allows the network to learn and adapt to the varying reliability of different annotators across different regions of the image.

The scalar reliability branch is implemented by taking the bottleneck features from the encoder and applying a global average pooling operation followed by a dense layer with sigmoid activation. This results in a single reliability value per annotator for each input image, making it computationally efficient but less spatially aware of reliability variations within the image.

5.1.3 Feature-based Reliability Estimation

The feature-based reliability estimation approach provides a more detailed spatial understanding of annotator reliability by operating on the bottleneck features of the encoder. This approach maintains the spatial information from the encoder's bottleneck while being more computationally efficient than pixel-wise reliability estimation.

The feature-based reliability branch is implemented by applying two convolutional layers with batch normalization to the bottleneck features:

- A first convolutional layer with 32 filters to process the bottleneck features
- A second convolutional layer that produces R feature maps, one for each annotator

This approach provides a coarse-grained spatial reliability map that captures regional variations in annotator reliability while maintaining computational efficiency. The resulting reliability maps have the same spatial dimensions as the bottleneck features, providing a middle ground between scalar and pixel-wise reliability estimation.

5.1.4 Pixel-wise Reliability Estimation

The pixel-wise reliability estimation approach provides the most detailed spatial understanding of annotator reliability by operating on the full-resolution features from the decoder. This approach allows for fine-grained reliability estimation at the pixel level, capturing local variations in annotator reliability.

The pixel-wise reliability branch is implemented by applying three convolutional layers with batch normalization to the decoder's output features:

- Two initial convolutional layers with 32 filters each to process the features
- A final convolutional layer that produces R reliability maps, one for each annotator

This approach provides the most detailed spatial reliability information, allowing the model to capture pixel-level variations in annotator reliability. The resulting reliability maps have the same spatial dimensions as the input image, providing the finest granularity of reliability estimation among the three approaches.

5.1.5 Integration with TGCE$_{SS}$ Loss

The model produces two distinct outputs: segmentation masks $\hat{\mathbf{Y}} = f(\mathbf{X};\theta)$ and reliability maps $\{\Lambda_r(\mathbf{X};\theta)\}_{r=1}^{R}$. These outputs work in tandem with the TGCE$_{SS}$ loss function described in Section 4.1. The loss function simultaneously guides the learning of both the segmentation masks and reliability maps, ensuring that the model learns to balance the contributions of different annotators based on their estimated reliability.

The TGCE$_{SS}$ loss function is implemented differently for each reliability approach, with specific adaptations to handle the different spatial dimensions of the reliability maps. All implementations share common hyperparameters:

- $q = 0.1$: Controls the truncation of the loss function
- $\epsilon = 10^{-8}$: Small constant to prevent numerical instability
- $\alpha_{reg} = 0.1$: Weight for the regularization term
- $\alpha_{entropy} = 0.1$: Weight for the entropy regularization
- $\alpha_{sum} = 0.1$: Weight for the sum-to-one constraint

For the scalar reliability approach, the loss function operates on global reliability values per annotator. The reliability maps are expanded to match the spatial dimensions of the predictions through tiling operations. This approach is computationally efficient but provides the same reliability value across all spatial locations.

The feature-based reliability approach requires additional processing to handle the spatial dimensions of the bottleneck features. The reliability maps are resized using bilinear interpolation to match the spatial dimensions of the predictions. This approach provides coarse-grained spatial reliability information while maintaining computational efficiency.

The pixel-wise reliability approach operates directly on full-resolution reliability maps, requiring no spatial transformations. This approach provides the most detailed spatial reliability information but requires more computational resources.

All three implementations include three regularization terms to ensure proper learning of the reliability maps:

- **Center Regularization**: $\alpha_{reg} \cdot \mathbb{E}[(\Lambda_r - 0.5)^2]$
- **Entropy Regularization**: $-\alpha_{entropy} \cdot \mathbb{E}[\Lambda_r \log(1+\Lambda_r) + (1-\Lambda_r)\log(1+(1-\Lambda_r))]$
- **Sum-to-One Regularization**: $\alpha_{sum} \cdot \mathbb{E}[(\sum_r \Lambda_r - 1)^2]$

These regularization terms help prevent the model from assigning extreme reliability values and ensure a balanced distribution of reliability across annotators. The loss function also includes a noise tolerance parameter ($\eta = 0.1$) to handle cases where the model's predictions differ from the annotators' labels.

The final loss function combines the main TGCE$_{SS}$ terms with the regularization terms:

$$\mathcal{L} = \mathbb{E}[\text{term}_1 + \text{term}_2] + \alpha_{reg} + \alpha_{entropy} + \alpha_{sum} \tag{5-1}$$

where term$_1$ and term$_2$ are the main TGCE$_{SS}$ terms that handle the interaction between predictions and reliability maps.

5.1.6 Training Process

The training process begins with the initialization of the ResNet-34 backbone using pre-trained weights, providing a strong foundation for feature extraction. The model's hyperparameters are optimized using Bayesian optimization through Keras Tuner, which searches for optimal values of:

- Learning rate: Sampled logarithmically between 10^{-5} and 10^{-2}
- TGCE$_{SS}$ parameters:
  - q: Controls truncation of the loss function (0.1 to 0.9)
  - Noise tolerance: Controls model's robustness to label noise (0.1 to 0.9)
  - Regularization weights: Control the balance between different loss terms (0.01 to 0.5)

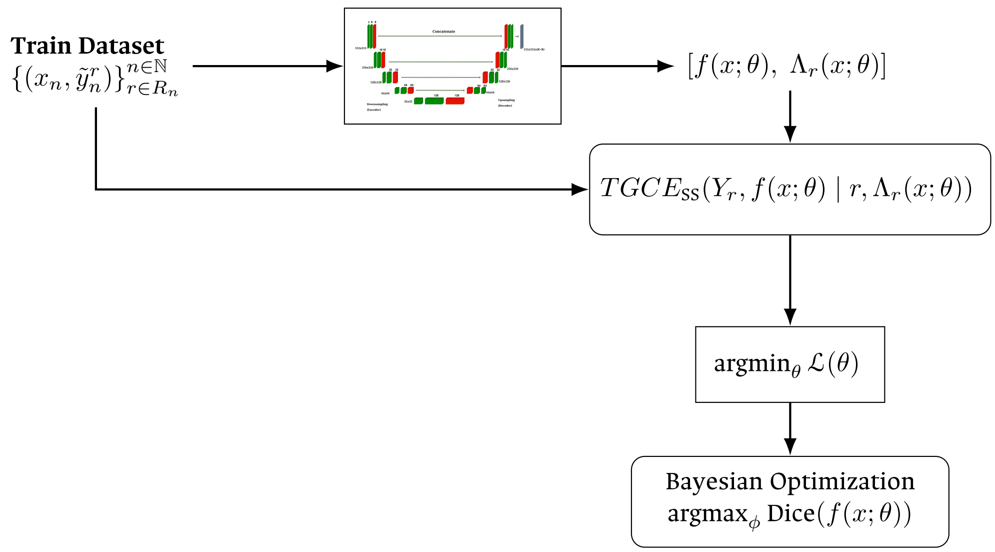


Figure **5-3** Training process of the proposed model overview. The model first undergoes hyperparameter optimization using Bayesian optimization, followed by end-to-end training with the optimal parameters. Estimated segmentation and reliability maps are computed for each input image, and the loss function is computed for the entire batch. Network parameters are tuned based on the loss function as objective (with minimization target) whereas the hyperparameters ϕ tuning are based on the Dice coefficient of the validation data set (with maximization target).

The hyperparameter search is performed using a validation set, with the objective of maximizing the Dice coefficient on the validation data. After finding the optimal hyperparameters, the model is trained end-to-end using the Adam optimizer with the best learning rate found. The TGCE$_{SS}$ loss function plays a crucial role in updating both the segmentation and reliability branches, ensuring that the model learns to effectively handle multiple annotators' inputs while accounting for their varying reliability.

The model's architecture is specifically designed to address the challenges of multi-annotator segmentation. Through the ResNet-34 backbone, it learns robust segmentation features that capture high-level patterns in the data. The UNET's skip connections enable the preservation of fine-grained details, while the parallel reliability branch allows the model to adapt to annotator-specific characteristics. This comprehensive design enables the model to effectively handle multiple annotators' inputs while maintaining high segmentation accuracy and reliability estimation.

5.2 Experimental Setup

5.2.1 Noisy Annotations Generation

A challenge arises for the creation of emulated noisy annotations datasets, since it is expected for images annotations to have some degree of expertise variability, similar to what is expected in real crowdsourced datasets. Simply introducing random noise into the annotated masks does not work, since the original morphological structures from the expected ground truth are far from being preserved. Instead, a "noisy" labeler is expected to produce an annotation which has at least some degree of morphological consistency on itself, even if it shows discrepancies when compared with some metric (like DICE score) against the ground truth annotation.

This has been proven experimentally when introducing random noise into any popular segmentation dataset, in which the resulting mask is just a non coherent map of noise across the image, as shown in Figure 5-4.

For this reason, a more sophisticated approach was needed to create datasets with emulated noisy annotations. The approach used here was to use a pre-trained U-Net model to produce a good enough [1] segmentation of the image, which was then used to produce a "noisy" annotation by introducing random noise into the last encoder layers of the model, thus preserving the morphological consistency of the original annotation. This strategy works since slight modifications introduced

[1] More precisely, a U-Net architecture with Resnet-34 backbone model with a simple categorical cross-entropy loss which reached 0.8426 Dice Coefficient after 44 epochs.

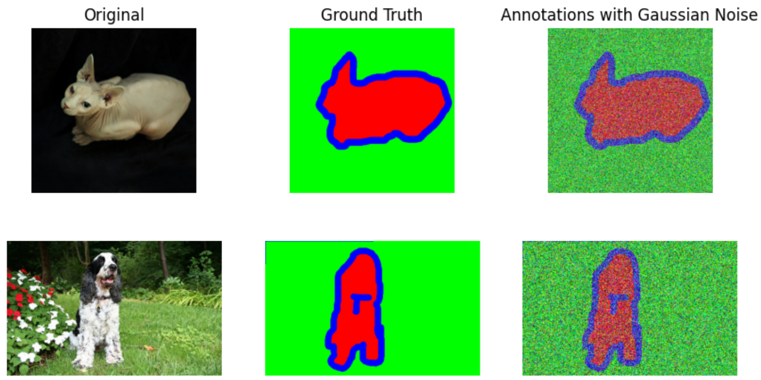


Figure **5-4** Example of a noisy mask generated by naively introducing random noise into a ground truth mask. Morphological consistency is lost.

into the encoder layers weights, resemble conceptual disturbances with respect to the respect to the original ground truth label, which kind of emulates the cultural behavior of having a different interpretation of the image, either for having a different level of expertise or for having a different point of view or school of thought. The first encoding layers were not modified, since it is expected human labelers would agree on the most fundamental structures (analog to extracted features from initial convolutional layers) in a similar way, thus preserving the morphological consistency of the original annotation.

In this way, the level of "disturbance" with respect to the ground truth for an emulated annotator can be controlled by the level of noise introduced into the weights of the last encoder layers, thus:

$$\mathbf{W}_{noisy} = \mathbf{W}_{original} + \epsilon_d \tag{5-2}$$

where $\mathbf{W}_{original}$ represents the original weights of the last encoder layers, ϵ_d follows a Gaussian distribution with zero mean and variance σ^2. The variance σ^2 controls the level of noise introduced, and thus the degree of disturbance in the resulting segmentation masks.

With the application of the encoder layer weight perturbation technique, the resulting noisy masks are shown in Figure 5-5. It can be seen that the morphological consistency is preserved, even though the resulting masks are far from the ground truth annotations, which goes perfectly well for testing the robustness of the models against noisy annotations in crowdsourcing-like scenarios.

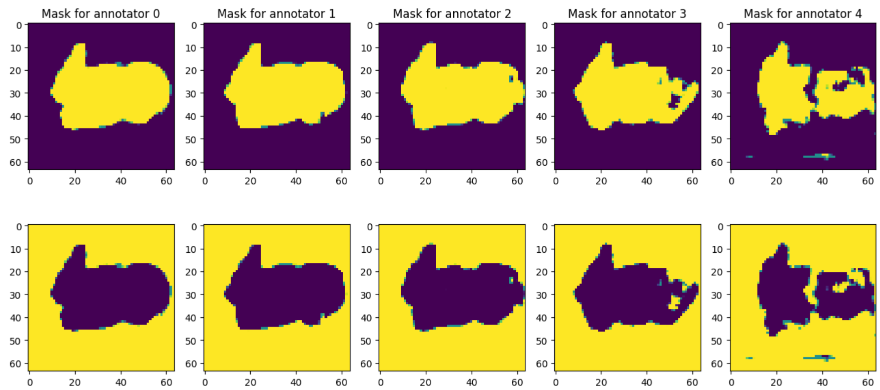


Figure **5-5** Noisy mask generated by enhancing the disturbances in the encoder layers weights for the Oxford-IIIT Pet Dataset. Morphological consistency is preserved. From left to right, SNR levels of noise in the encoder layer are 10, 5, 2, 0, -5 dB. Not adding noisy to the encoder layer would reach to an annotator behavior as good as the originally trained base model.

5.2.2 Missing Labels Emulation

It has also been determined that most real world crowdsourced datasets suffer from missing labels, since not all annotators are able to label all of the input samples. To emulate this behaviour, a missing labels dataset was created by randomly skipping labels from the annotators labels based on a labeling rate $r_l \in [0, 1]$. The labeling rate r_l controls the proportion of labels that are skipped from the annotators labels. The lower the labeling rate, the more labels are skipped from the annotators labels. The implementation of this missing labels emulation together with the noisy annotators generation can be flexibilized through an algorithm, which enables it to be used in experiments with variable amount of labelers, noise levels, and labeling rates. An overview of the algorithm is shown in Algorithm 5-1. A sample of the generated noisy and missing labels dataset is shown in Figure 5-6.

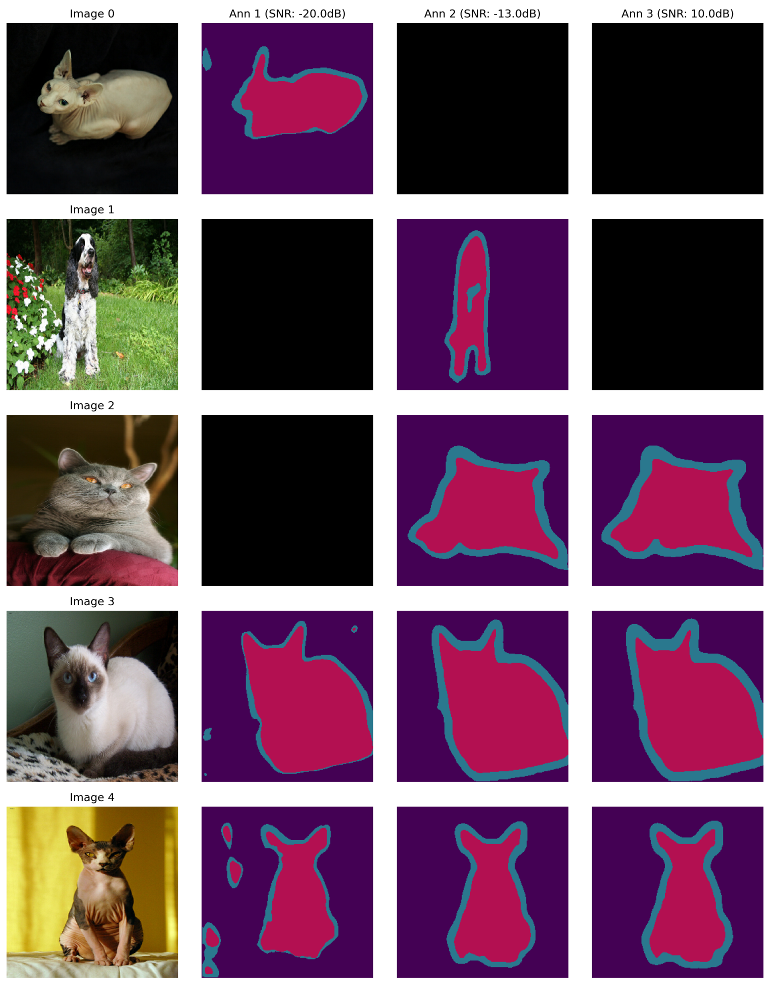


Figure **5-6** Example of missing labels and noisy annotations emulation for the Oxford-IIIT Pet Dataset. The same label rate ($r_l = 0.7$) is preserved for all labelers, which means in this case that every labeler is expected to label 70% of the samples, and skip the remaining 30%.

Algorithm 5-1 Multi-Annotator Dataset Generation with Noisy and Missing Labels

Require: Base segmentation model M, SNR levels $\{s_1, ..., s_n\}$, labeling rate $r \in [0, 1]$
Ensure: Dataset with multiple noisy annotations and missing labels

Step 1: Generate Noisy Annotators
for each SNR level s_i **do**
    Create disturbed model M_i by adding noise to M with SNR s_i
end for
Step 2: Initialize Labeler Assignment System
Create assignment matrix $A \in \{0, 1\}^{N \times n}$ where N is number of samples
for each annotator $j \in \{1, ..., n\}$ **do**
    for each sample $i \in \{1, ..., N\}$ **do**
        $A_{i,j} \leftarrow \begin{cases} 1 & \text{with probability } r \\ 0 & \text{with probability } 1 - r \end{cases}$
    end for
end for
Step 3: Generate Multi-Annotator Dataset
for each image x in dataset **do**
    Resize x to target dimensions
    for each annotator model M_i **do**
        Generate annotation $y_i = M_i(x)$
        Apply labeler assignment mask: $y_i \leftarrow y_i \cdot A_{i,j}$
    end for
    Store $(x, \{y_1, ..., y_n\})$ in dataset
end for
return Multi-annotator dataset with noisy and missing labels

5.3 Results

5.3.1 Oxford-IIIT Pet Dataset

The experimental results on the Oxford-IIIT Pet Dataset demonstrate interesting insights into the performance of different approaches. While STAPLE achieves the highest Dice and Jaccard scores (0.8718 and 0.7917 respectively) for the dataset with no missing labels (label rate of 1.0), our proposed TGCE$_{SS}$ approaches show superior performance for label rates below 1.0 (Table 5-1). The pixel-wise variant, while slightly lower in performance metrics, provides more detailed spatial reliability information. It is of note that the key strength of TGCE$_{SS}$ lies in its ability to simultaneously learn both segmentation and reliability estimation, providing valuable insights into annotator behavior and confidence levels that STAPLE and MV cannot offer.

Method	**Label Rate 1.0**		**Label Rate 0.7**		**Label Rate 0.3**	
	Dice	Jaccard	Dice	Jaccard	Dice	Jaccard
TGCE scalar	0.8622±0.009	0.7791±0.012	**0.8568±0.017**	**0.7726±0.024**	**0.8529±0.004**	**0.7677±0.007**
TGCE feature	0.6995±0.044	0.5733±0.047	0.4766±0.026	0.3441±0.027	0.4766±0.026	0.3441±0.027
TGCE pixel	0.8468±0.018	0.7589±0.025	0.8545±0.013	0.7695±0.017	0.8283±0.001	0.7359±0.002
MV	0.8132±0	0.7222±0	0.7572±0	0.6554±0	0.5566±0	0.4687±0
STAPLE	**0.8718±0**	**0.7917±0**	0.8439±0	0.7609±0	0.5734±0	0.5158±0

Table **5-1** Performance metrics (Dice and Jaccard) at different labeling rates. Best values per category are bolded.

The reliability estimation capabilities of our model are particularly noteworthy, as demonstrated in Figures 5-7 and 5-8. The model successfully captures the varying reliability levels of different annotators, with higher reliability assigned to annotators with lower noise levels (10 dB) and lower reliability to those with higher noise levels (-20 dB). This ability to quantify and visualize annotator reliability is a significant advantage over traditional approaches like STAPLE and MV, which focus solely on segmentation accuracy without providing insights into annotator behavior.

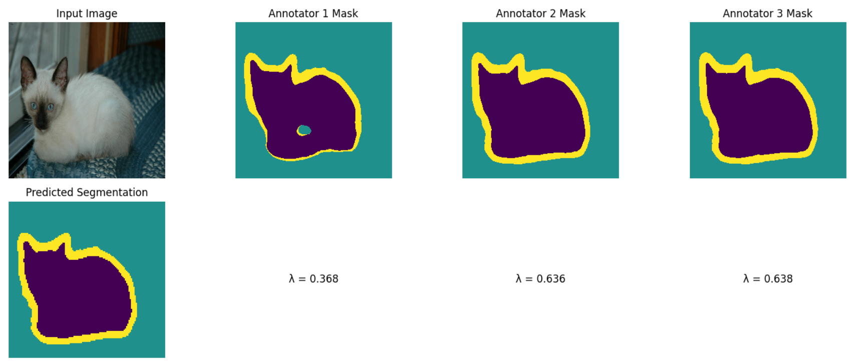


Figure **5-7** Example evaluation of the proposed model in its last trained epoch. An accurate segmentation output is delivered as well as scalar reliability estimations from the labelers masks. Disturbance signal-to-noise ratios are: $[-20, 0, 10]$ dB from left to right. Best found hyperparameters: learning rate = 0.0011, $q = 0.4$, noise tolerance = 0.2, $\alpha_{reg} = 0.26$, $\alpha_{entropy} = 0.03$, $\alpha_{sum} = 0.27$.

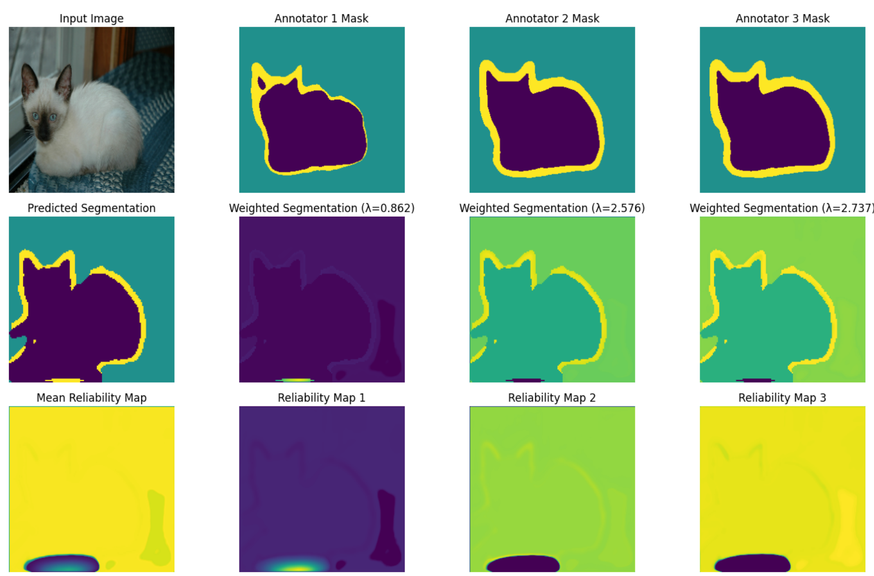


Figure **5-8** Example evaluation of the proposed model in some given epoch (5/50) of training, which demonstrates a quick convergence of the model for correctly estimating the reliability of the labelers masks. An accurate segmentation output is delivered as well as pixel-wise reliability estimations from the labelers masks. Disturbance signal-to-noise ratios are: $[-20, 0, 10]$ dB from left to right. Best found hyperparameters: learning rate = 0.0027, $q = 0.9$, noise tolerance = 0.2, $\alpha_{reg} = 0.38$, $\alpha_{entropy} = 0.02$, $\alpha_{sum} = 0.42$.

5.3.2 TNBC Dataset

As mentioned in Section 2.4.2, [Amgad et al., 2019] presented a Triple Negative Breast Cancer (TNBC) dataset, which is crowdsourced and available as HistomicsTK metadata. Later, [López-Pérez et al., 2023] sampled 151 Whole Slide Imagings (WSIs), from which 161 Region of Interests (ROIs) were extracted and labeled by 20 medical students. The dataset was sampled as patches of size 512x512, resulting in 10173 training patches, 1264 validation patches and 399 testing patches. The segmentation masks for the patches are composed of the classes: tumor, stroma, inflammation, necrosis, and others [2]. This subset of the TNBC is named Crowd-Seg.

Raw PNG files for the images patches and masks were provided by the authors, in practice, in order to optimize the use of these data, a custom tensorflow dataset was built for facilitating batch memory, caching and prefetching. The custom dataset implementation handles the loading and preprocessing of both images and masks, including normalization of images to the [0,1] range and one-hot encoding of the segmentation masks. It also manages the multi-annotator structure by creating a labeler mask that indicates which annotators provided labels for each sample. The dataset supports data augmentation through random transformations (flips, rotations, brightness/contrast adjustments) and includes visualization utilities for debugging and analysis purposes. This custom dataset implementation is publicly available as a PyPI package (see Section 6.3).

Notes on the TNBC Dataset

The TNBC dataset contains a couple of peculiarities that are worth mentioning. First, the label rate (inverse percentage of labels that are missing) is not uniform across the labelers and, in general, quite low, ranging from 1 to 12% as seen in Figure 5-9.

[2] In practice, some of this masks also contain an additional class labeled as ‘ignore’.

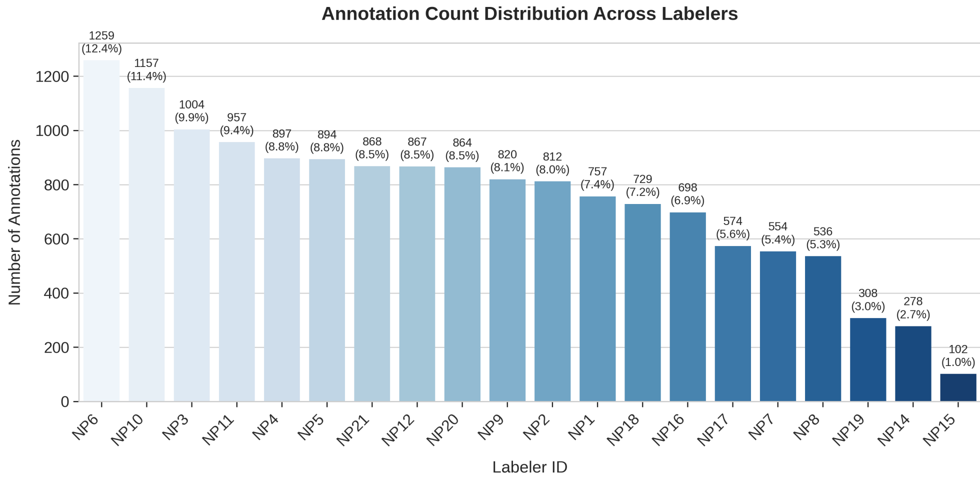


Figure **5-9** Label rate distribution across the labelers for the TNBC (Crowd-Seg) dataset.

On the other hand, the classes distribution across all patches are not equal, being 'Tumor' (class 2) the most frequent class. Despite the classes not being equally distributed, the distribution of the classes across the labelers is fairly uniform, as shown in Figure **5-10**.

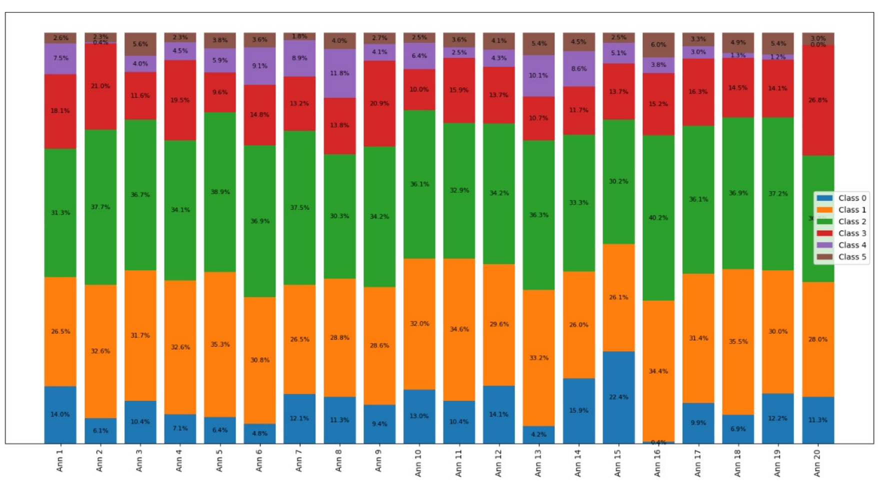


Figure **5-10** Classes distribution across the labelers for the TNBC (Crowd-Seg) dataset. Most common classes are 'Tumor' (class 2),'Other' (class 1) and 'Stroma' (class 3).

It was also found that the dice score of the labelers masks against the ground truth differed across the labelers and the classes, as shown in Figure **5-11**. The target class (Tumor) ranged dice scores across the labelers from 0.5 to 0.9, other classes can get as low as 0.4 for in average for some labelers, meaning very high levels of disagreement.

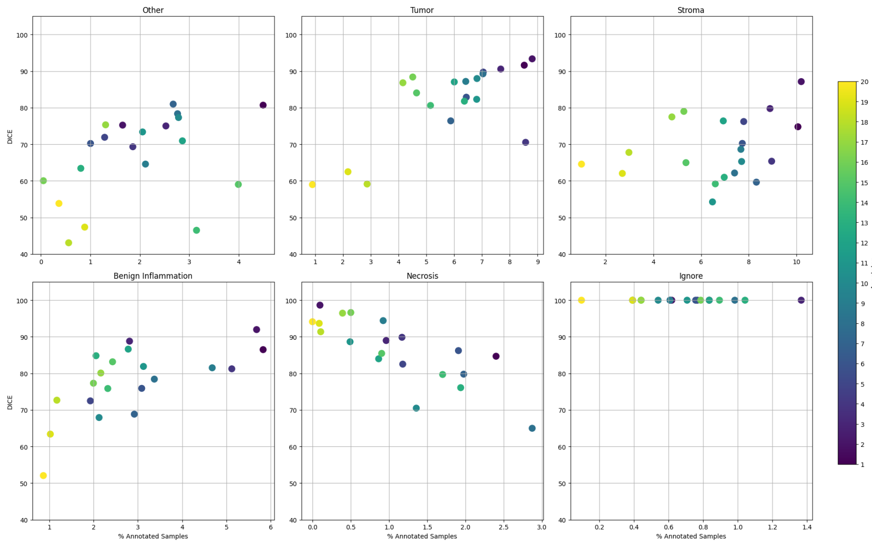


Figure **5-11** Dice score of the labelers masks against the ground truth for the TNBC (CrowdSeg) dataset.

Finally, it was found that the expert labels, which would thereby be considered as ground truth for metrics evaluation, were not always correct, as shown in Figure 5-12. These morphological inconsistencies make an objetive metric for evaluating the performance of the proposed model difficult to define, since the ground truth is not always correct or expressive enough.

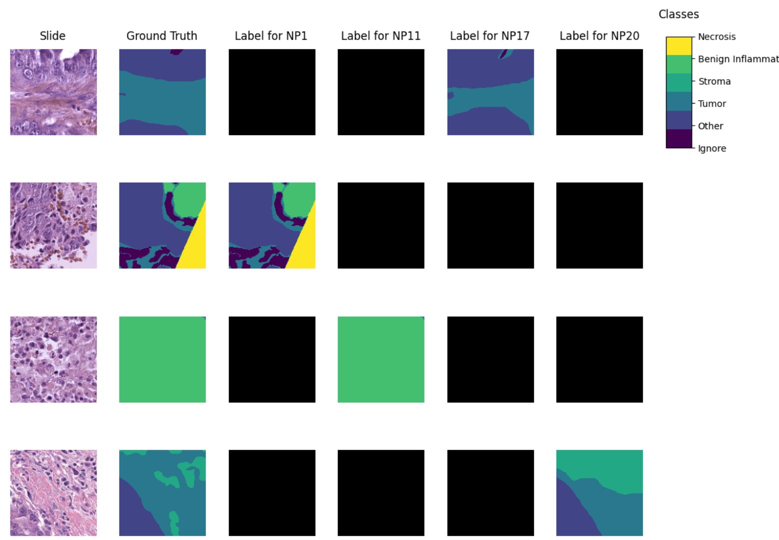


Figure **5-12** Example of ground truth with odd geometries (second patch from top to bottom), which are not always correct or compatible with the underlying structures of the patch itself.

Special considerations for the TNBC Dataset

Because of the aforementioned peculiarities of the TNBC dataset, some special considerations were taken into account for the experiments. Because of the low label rates, experiments showed an unavoidable convergence to estimate arbitrarily high reliabilities on labelers which did not label a certain patch, which is not a desired behavior. None of the terms from the proposed loss function penalizes this optimization shortcut, so to avoid this, an additional regularizer was included for penalizing high estimated reliabilities on unlabeled data, as follows:

$$\mathcal{L}_\lambda = \max\left(0, \sum_{\forall i \in R} \frac{1}{1-\alpha} \left(\Lambda_i - \mathbb{LM}_i\right) \cdot e^{\frac{\Lambda_i - \mathbb{LM}_i - 1}{\beta}}\right) \tag{5-3}$$

where R is the amount of labelers, Λ_i is the estimated reliability of the labeler i for a certain patch and $\mathbb{LM}_i$ is the one hot label mask of the labeler i for the same patch.

The α and β parameters control the penalty strength for high and low estimated reliabilities, respectively. In the Crowd-Seg experiments, this regularizer replaced the other regularizers, since it was showing better convergence on estimating the reliability of the labelers. This leads to an special overall loss function for the TNBC dataset of the form:

$$\mathcal{L} = \mathcal{L}_{GCE} + \gamma \mathcal{L}_{\lambda}$$

where γ is a hyperparameter that controls the trade-off between the GCE loss and the regularizer.

Figure **5-13** Training curves for the baseline model on the TNBC (Crowd-Seg) dataset.

5.4 Results

Initially, a baseline model (ground truth only) was trained with an early stopping mechanism, which stopped the training when the validation loss stopped decreasing for 10 epochs, reaching only 20 epochs of significant training. The results are shown in Figure **5-14**.

Then, using the scalar model described in Section 5.1.2, training was performed for 20 epochs and Bayesian hyperparameter optimization on learning rate, decoder dropout rate, α and β for the regularizer, and q and noise tolerance from Generalized Cross Entropy (GCE). The results are shown in Figure **5-15**.

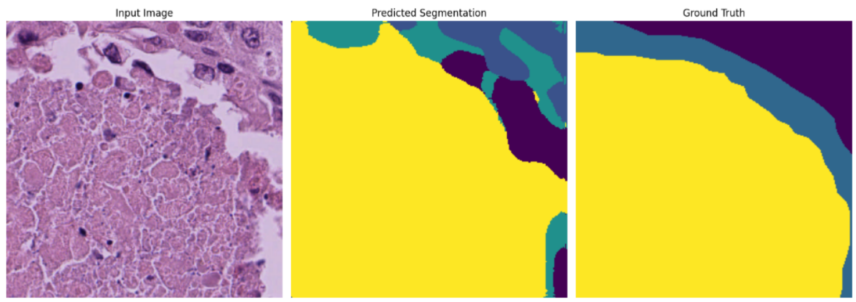


Figure **5-14** Segmentation results for baseline model on the TNBC (Crowd-Seg) dataset after 20 epochs of training. Even though the segmentation algorithm seems to show progress on its way to capture the underlying target structures ot be segmented, since the ground truth annotation is not expressive enough, the Dice coefficient remains below 0.55, which is not considered a good performance. The baseline model remained with overfitting even after combining techniques of: Bayesian hyperparameter optimization, data augmentation, learning rate scheduling, dataset shuffling, and early stopping.

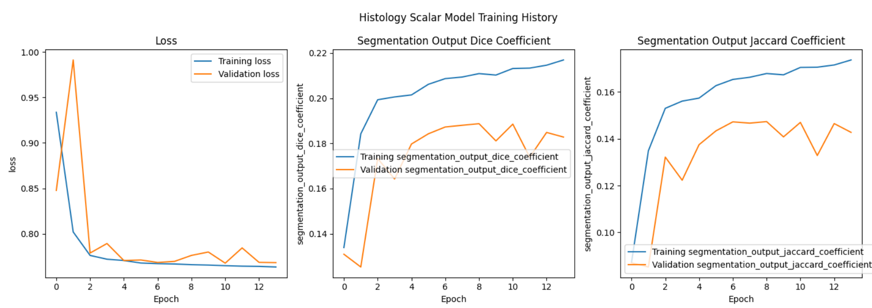


Figure **5-15** Segmentation results for scalar model on the TNBC (Crowd-Seg) dataset after 20 epochs of training. Again, training metrics show a constant grow, but not validation metrics, which shows a combined issue of overfitting and unreliable ground truth.

5.5 Conclusion

The experimental results on both the Oxford-IIIT Pet and TNBC (Crowd-Seg) datasets provide valuable insights into the performance and limitations of our proposed approach. The model demonstrated exceptional performance on the Oxford-IIIT Pet dataset, achieving competitive Dice and Jaccard scores while successfully capturing annotator reliability patterns. The model's ability to simultaneously learn segmentation and reliability estimation proved particularly valuable, providing insights into annotator behavior that traditional approaches like STAPLE and MV cannot offer.

However, the experiments on the TNBC dataset revealed significant challenges in evaluating semantic segmentation models for histopathological images when dealing with unreliable ground truth. The morphological inconsistencies in the expert annotations, combined with the extremely low label rates (1-12%) across annotators, made it difficult to objectively measure the model's performance. While the model showed promising convergence in training, the validation metrics remained unstable, suggesting that the current dataset may not be suitable for rigorous evaluation of multi-annotator segmentation approaches.

These findings highlight the need for more reliable ground truth annotations in histopathological image segmentation datasets. Future work should focus on:

- Developing datasets with more consistent and reliable expert annotations
- Implementing stricter quality control measures for ground truth generation
- Exploring alternative evaluation metrics that account for annotation uncertainty
- Investigating methods to handle extremely sparse annotations more effectively

Despite these challenges, the model's performance on the Oxford-IIIT Pet dataset demonstrates its effectiveness in handling multiple annotators with varying reliability levels. The model successfully learned to:

- Balance the contributions of different annotators based on their reliability
- Provide detailed spatial reliability information through pixel-wise estimation
- Maintain high segmentation accuracy even with missing labels

- Adapt to different noise levels in annotator inputs

These capabilities make the proposed approach particularly valuable for real-world applications where multiple annotators with varying expertise levels contribute to the dataset. The model's ability to simultaneously learn segmentation and reliability estimation provides a more comprehensive understanding of the annotation process, which could be leveraged to improve the quality of future annotations and dataset construction.

CHAPTER

SIX

FINAL REMARKS

6.1 Conclusions

This thesis has presented a novel approach to address the challenges of Image Semantic Segmentation (ISS) in histopathology images, particularly in crowdsourcing scenarios where multiple annotators with varying levels of expertise contribute to the labeling process. The proposed solution successfully tackles the fundamental challenges of modeling annotator performance and handling label inconsistencies while maintaining computational efficiency. The work demonstrates significant improvements over existing state-of-the-art methods, particularly in scenarios with noisy and missing labels, which are common in crowdsourcing approaches.

Throughout this work, several techniques have been explored to develop a model that can face Image Semantic Segmentation (ISS) of real world histopathology images in crowdsourcing scenarios. Different approaches were studied and evaluated to reach an optimal solution which could model the annotators' performance from the input space. The following remarks are based on the results obtained from the experiments conducted and reviewed literature:

- Chained Gaussian Processes approaches, while robust for classification and regression tasks, proved computationally intensive for image segmentation. This limitation highlighted the need for more efficient approaches that could handle the complexity of medical image segmentation while modeling annotator performance.

- The development of a novel loss function was crucial for the success of this work, as existing loss functions were insufficient to model the annotators' performance across the input space. The proposed loss function successfully enabled the model to adaptively infer the best possible segmentation without requiring separate inputs about labeler performance.

- The introduction of a tensor map to codify labeler reliability proved to be an effective approach, allowing the model to implicitly weigh the labelers based on their performance across the mask and classes space, thus handling the labeler performance inconsistency effectively.

- The use of U-Shaped Deep Learning models as a building block for the proposed solution was validated, as it effectively captured spatial relationships between pixels and annotations while handling the variability in image quality.

- The combination of Chained Deep Learning approaches with the proposed loss function and tensor map representation resulted in a comprehensive solution being, to the best of our knowledge, the first approach on offering an image segmentation solution for multiple labelers in medical image with interpretability in the reliability of the labelers, successfully addressing both labeler inconsistency and image quality variability. The model outperformed other state of the art solutions in terms of segmentation accuracy in noisy and missing labels conditions, which is rather common in crowdsourcing approaches.

6.2 Future work

As tools and frameworks for bayesian approaches like Chained Gaussian Processes [1] evolve, it is expected to see a decrease in computational requirements and hence, making mixed models more appealing for the task of ISS of histopathology images. In that sense, the use of Correlated Chained Gaussian Processes for Multiple Annotators (CCGPMA) as a building block for the proposed solution could be a good approach, as it is a powerful model that can take advantage of the inter-dependencies between the annotators, and the powerfulness of CNNs to extract features from the images.

The noisy labelers emulation technique developed in this work presents an opportunity to test the model's robustness in various medical image segmentation challenges. This technique could be adapted to simulate different types of expert variability in other medical imaging domains:

- **Multi-modal Medical Imaging**: The technique could be extended to simulate expert disagreements in multi-modal scenarios, where different imaging modalities (e.g., MRI, CT, PET) might lead to varying interpretations of the same anatomical structure.
- **Temporal Consistency**: For longitudinal studies, the emulation could be modified to simulate how expert annotations might vary over time, helping to test the model's ability to maintain consistency in segmentation across different time points.
- **Cross-institutional Variability**: The noise injection technique could be calibrated to simulate the variations in annotation patterns that arise from different institutional protocols or training backgrounds.

[1] Including GPFlow, and GPFlux.

- **Pathology-specific Challenges**: The emulation could be tailored to simulate expert disagreements in specific pathological conditions, where the morphological features might be more ambiguous or subject to interpretation.

These extensions would help validate the model's performance across a broader range of medical imaging scenarios and ensure its robustness in real-world clinical applications.

The pixel-wise reliability mapping approach shows particular promise for future development. While the current implementation already demonstrates good performance, several enhancements could lead to significant improvements:

- **Advanced Data Augmentation**: Implementing a comprehensive data augmentation strategy could significantly improve the model's convergence and performance. This could include:
  - Geometric transformations (rotations, flips, elastic deformations)
  - Intensity variations to simulate different staining conditions
  - Multi-scale training to handle varying tissue sizes
  - Adversarial training to improve robustness

  These augmentations would help the model learn more robust features and better generalize to unseen data.
- **Attention Mechanisms**: Integrating attention mechanisms into the pixel-wise reliability branch could help the model focus on the most relevant regions for reliability estimation, potentially improving both accuracy and computational efficiency.
- **Multi-scale Reliability Estimation**: Implementing a hierarchical approach to reliability estimation could capture both fine-grained local variations and broader regional patterns in annotator reliability, this would imply a network with yet more branches and outputs.

The combination of these enhancements with the existing pixel-wise reliability mapping approach could potentially outperform all other techniques, as it would provide the most detailed and accurate representation of annotator reliability while maintaining computational efficiency. The use of data augmentation, in particular, could help the model converge faster and achieve better generalization, as it would expose the model to a wider variety of possible variations in both the input images and the corresponding annotations.

6.3 Academic Discussion

To facilitate the adoption and further development of the proposed approach, several resources have been made publicly available. A comprehensive GitHub repository [2] has been created that provides easy-to-use APIs for implementing the proposed solution. The repository includes pre-built dataset loaders for TensorFlow/Keras frameworks, implementation of the TGCE_{SS} loss function in all its variants (scalar, features, and pixel-wise), and a neural network architecture builder with experimental examples. The two *tfds* datasets available in the repository are:

- **Triple Negative Breast Cancer (TNBC) Dataset**: A dataset of histopathology images of breast cancer, with annotations of the tumor regions from [López-Pérez et al., 2024]. The dataset is automatically fetched from AWS S3, which eliminates the need for manual downloads or external dependencies.
- **Oxford-IIIT Pet Dataset**: A dataset of pet images, based on the well-known Oxford-IIIT Pet Dataset with annotations of the pet regions. The mechanism for emulating noisy crowdsourced annotations exposed throughout the present work (see Section 5.2.1) is also available in the repository in a tunable manner, just as the TNBC dataset, this `tfds` is batteries included

[2] https://github.com/blotero/seg_tgce

(automatic fetching and mapping, no need to manually download any external source).

To further simplify the adoption of the proposed solution, a PyPI package has been published with detailed documentation available at https://seg-tgce.readthedocs.io/en/latest/. The documentation provides comprehensive installation instructions, API reference, usage examples, and best practices for implementation. This package makes it straightforward for the community to integrate the proposed approach into their existing workflows, with a focus on ease of use and clear documentation of all available features and options.

An academic publication is currently being prepared for submission to a Q1 journal. This publication will provide detailed theoretical foundations of the proposed approach, comprehensive experimental results and analysis, comparison with state-of-the-art methods, and discussion of limitations and future directions. The publication will serve as the primary reference for the theoretical and practical aspects of the proposed solution, ensuring that the work is properly documented and validated through the peer-review process.

These resources collectively contribute to the reproducibility and accessibility of the research, enabling the broader scientific community to build upon and extend the proposed approach. The combination of open-source code, well-documented APIs, and peer-reviewed publication ensures that the work can be properly evaluated, validated, and utilized by other researchers in the field. This comprehensive approach to sharing research outputs aligns with current best practices in scientific computing and Machine Learning research, promoting both transparency and practical utility of the proposed solution.

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